

Master Thesis

Multi agent systems for image analysis: object and complex scene detection

Cross-Modal Agent Disagreement as a Bounded
Signal for Epistemic Uncertainty in Low-Fidelity
Historical Scans

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Abstract

The automated analysis of domain-shifted visual collections (such as historical archives, medical imagery, satellite scans, and archaeological databases) is fundamentally constrained by the irreducible semantic ambiguity of low-fidelity or out-of-distribution content. While modern single-model architectures achieve high predictive accuracy, they are structurally incapable of diagnosing their own epistemic uncertainty. We present a *Dual-Core Validation and Multi-Agent Curation (DCV-MAC)* framework that provides an operational, empirical proxy for epistemic uncertainty in complex visual collections through cross-modal model disagreement – not a direct measurement of a model’s true epistemic state. Our central contribution demonstrates that the **fusion uncertainty** derived from cross-modal agent disagreement between a spatial object detector (YOLO) and a semantic vision-language model (LLaVA) is a highly discriminative proxy for prediction errors on ambiguous, scene-positive content: the SAA uncertainty score achieves an error-detection ROC-AUC of **0.9724** on the $n = 801$ human-verified gold cohort (vs. 0.6545 for temperature-scaled single-model entropy). A group-level (publication-disjoint) held-out re-evaluation refines this claim: disagreement remains highly discriminative on positive, scene-containing images (ROC-AUC = 0.980), but is anti-correlated with error on negative-control scans where all agents share a common false-positive bias (ROC-AUC = 0.020) — a concrete instance of correlated failure that bounds the scope of the disagreement signal rather than invalidating it. We further formalize this signal in a distribution-free **Conformal Risk Control** triage policy: calibrated and tested on the same positive-heavy C_1 cohort, it appears to certify automatic approval of **87.5%** of the collection while capturing **94.3%** of true human classification errors in the review queue (a **7.57×** efficiency lift, formally bounded test risk $\mathcal{R}(\hat{\lambda}) = 0.0075 \leq \alpha = 0.05$). Re-running the identical, unmodified procedure on a group-level (publication-disjoint), negative-control-inclusive calibration/test partition collapses this to a **0.98×** (chance-level) lift, with the formal risk bound **violated** ($\mathcal{R}(\hat{\lambda}) = 0.182$, exceeding every target α tested up to 0.15). We report this as the thesis’s central operational finding: disagreement-based triage — whether a simple heuristic or a formally certified conformal policy — is only trustworthy within the positive-heavy regime it is calibrated on, because that regime is precisely where agent disagreement tracks error; outside it, on unanimous

false positives, the signal is silent by construction. Critically, while the multi-agent fusion score achieves statistically equivalent raw scene-labelling accuracy to a monolithic baseline (Wilcoxon $p = 0.375$, Cohen’s $d = +0.07$, negligible), it provides a structurally superior mechanism for uncertainty routing *within its validated scope*: after post-hoc Platt scaling (logistic calibration fitted on a 50% held-out split), the fusion score achieves $\text{ECE} < 0.001$ on C_1 (reduced from $\text{ECE} = 0.75$ pre-calibration), compared to $\text{ECE} = 0.28$ for the uncalibrated monolithic baseline. Downstream from this validation core, auxiliary curation agents (Temporal, Contradiction, Bayesian Ranker, Cross-Archive Linker, and Grounded OCR Document) extract queryable historical claims without adding noise to the SAA consensus. Evaluated against a human-verified gold subset ($n = 801$), our system maintains a Triage Precision@10% of 1.000 on that cohort and demonstrates superior error calibration compared to single-model local VLM entropy baselines, at zero marginal API cost for the local validation core (Tiers 1–2; Tier 3’s optional frontier-API escalation is used only for a small ambiguous residual, at real per-call cost). A genuine cross-institutional pilot on $n = 24$ images downloaded live from the Europeana Open Culture API (10 real GLAM institutions; a prior $n = 30$ figure was found during a reproducibility audit to have used mislabeled in-archive Colibri images, not Europeana data, and has been withdrawn) shows the SAA uncertainty signal responds substantially to genuine domain shift (mean fusion score 0.332 vs. the Colibri baseline’s 0.762); a full transfer-of-generalization claim additionally requires human ground-truth annotation on this new sample, which is in progress. Ultimately, this work reframes the role of AI in digital humanities and broader computer vision not as an infallible oracle, but as a domain-agnostic, uncertainty-aware consensus network whose operating envelope must be measured and disclosed, not assumed, and which defers to human expertise precisely where automated analysis is least reliable and least able to know it.

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Chapter 1

Introduction

1.1 The Archival Challenge in the Digital Age

Historical illustrations are critical primary sources for historians, sociologists, and cultural researchers. They capture moments of social change, architectural evolution, and daily life that are nowhere else documented in such visual detail.

However, the sheer volume of digitized archives makes manual cataloging impossible. In the Colibri Historical Image Collection, which contains over 53,000 images, the lack of dual-modality (visual and textual) searchability significantly hampers the discovery of specific historical events and figures.

1.2 Identification of the Research Gap

While modern object detection has reached human-level performance on benchmarks like COCO [12] or OpenImages [10], these models rely on “High-Fidelity” samples. Historical photographs exist in a “Low-Fidelity” regime characterised by film grain, chemical decay, fading, and monochromatic capture. Consequently, modern detectors exhibit severe performance degradation on archival scans.

A natural response is to build a restoration-and-distillation pipeline: generate pseudo-labels with open-vocabulary teacher models (e.g., GroundingDINO) and distil them into a fast student detector (e.g., YOLOv11). We constructed this pipeline, achieving a seemingly impressive proxy mAP50 of 0.989. However, evaluating a student against synthetic pseudo-labels is a known methodological vulnerability; proxy metrics often mask deep epistemic failures in historical domains. Indeed, when deployed across all 12,110 Colibri images, the system exposed a severe structural limitation: a single model cannot diag-

nose its own hallucinations. Even with perfect proxy scores, a standalone model cannot tell you *when* it is failing on an ambiguous historical document, only that it has made a prediction.

This observation reframed the research question. Rather than asking “how do we make one model detect better?”, we asked: “**can the disagreement between models be used as a systematic signal for identifying where AI understanding breaks down on historical images?**” This is the central question of the *Dual-Core Validation and Multi-Agent Curation (DCV-MAC)* framework. The restoration pipeline does not disappear from this story — it is the infrastructure that normalises archival images into a visual regime where VLMs can produce comparable outputs. But the primary contribution is the discovery that **inter-agent disagreement is an epistemic signal**, not noise to be minimised.

The primary research gap addressed in this work is therefore twofold: (1) the lack of a restorative preprocessing layer that can bring archival images into alignment with modern VLM input distributions, and (2) the absence of a principled, interpretable mechanism for quantifying *how confident* automated systems should be when analysing semantically ambiguous historical content.

1.3 Generalization Beyond Historical Archives: A Domain-Agnostic Paradigm

While the primary empirical focus of this thesis is the Colibri historical image collection, the core architectural principles of the Dual-Core Validation and Multi-Agent Curation (DCV-MAC) framework are fundamentally domain-agnostic. The challenge of deploying machine learning models under severe domain shift—where the model’s confidence scores become uncalibrated and hallucinations go undetected—is a critical bottleneck across several domains in computer vision. Specifically, the DCV-MAC framework is directly applicable to:

- **Medical Imaging:** Where models trained on clean clinical benchmarks face domain shift when deployed in resource-limited clinics, making the calibration of epistemic uncertainty essential for patient safety.
- **Satellite and Aerial Imagery:** Where seasonal changes, atmospheric occlusion, and varying sensor qualities degrade the reliability of object detectors, necessitating

a cross-modal consensus layer to flag ambiguous regions.

- **Archaeology and Cultural Heritage:** Where visual artifacts, degraded carvings, and museum collections display stylized and fragmented features that deviate from standard photographic pre-training distributions.
- **Legal Evidence and Document Archives:** Where degraded scans, handwritten annotations, and complex layouts create visual-semantic conflicts that require multi-modal disagreement tracking to prevent automated misinterpretation.

By modeling uncertainty as inter-agent disagreement rather than relying on standalone confidence scores, our framework provides a general-purpose safety net for any low-fidelity visual regime.

1.4 Research Objectives and Goals

The overarching goal of this research is to bridge the "archival semantic gap" through a reproducible **multi-agent** framework for historical indexing and validation. By establishing the *Visual Historian-MAS* platform, we transform raw legacy scans into evidence-rich semantic records. Specifically, this work addresses the following research questions:

- **RQ1 – Uncertainty from Disagreement:** Can cross-modal agent disagreement reliably estimate epistemic uncertainty on domain-shifted visual collections?
- **RQ2 – Workload Reduction:** Does the multi-agent framework reduce human curation workload without sacrificing error recall?
- **RQ3 – Contradiction Detection:** Does cross-modal contradiction detection improve curation precision beyond Agent 0 alone?
- **RQ4 – Retrieval & Cross-Archive Discovery:** Can the framework support archival retrieval and cross-collection discovery?

1.5 Evaluation Methodology for Multi-Agent Analysis

The evaluation protocol is designed to isolate both predictive quality and validation quality:

- **Single-Agent Baselines:** Evaluate each agent independently (object detector, scene classifier, VLM reasoning, document/OCR) using standard metrics for its task.
- **Coordinator Evaluation:** Compare weighted multi-agent fusion against single-agent baselines using agreement, realism score distributions, and downstream indexing quality.
- **Ablation Analysis:** Remove one agent at a time and measure degradation in agreement, realism confidence, and review precision.
- **Complex-Scene Stratification:** Report performance separately for high clutter, low contrast, and high-occlusion subsets.
- **HITL Efficiency:** Quantify precision@k of disagreement-based triage and estimate reviewer effort reduction against random sampling.

1.5.1 Evaluation Cohort Reference Summary

To ensure methodological clarity and reproducibility across evaluation sections, Table 1.1 summarizes the primary evaluation cohorts used throughout this thesis, detailing their sample sizes, composition, analytical purpose, and associated experiments.

Table 1.1: Master Evaluation Cohort Summary across all experimental sections.

Cohort Symbol & Name	Sample Size (n)	Composition & Data Source	Primary Analytical Purpose	Associated Experiments
C_0 : Deployment Corpus	12,110	Full unlabelled institutional digitisation subset from the Colibri archive	Pipeline deployment, retrieval indexing, and scaling evaluation	Exp 1–6, Exp 27
C_1 : Expert Reference	801	796 scene-positive + 5 negative control images (CVAT expert-reviewed)	Primary calibration, AWLF error retraining, and HITL triage evaluation	Exp 20, Exp 29, Exp 30
C_2 : Spatial Audit	300	Single-rater spatial bounding box audit cohort (scene labels, box counts, confidence/clarity ratings)	Spatial error prediction and meta-learning	Exp 1 (Audit), Exp 31
C_3 : System Evaluation	2,000	801 C_1 expert images + 1,199 negative control catalog records	Source pool for the group-level train/calibration/test partition (Sec. 4.2) and system-level agent leave-one-out ablations	Exp 20 (Full), System Ablations
C_4 : Frontier Comparison	100	Stratified subset representing diverse archival categories	Monolithic baseline comparison (LLaVA-OV) and hybrid routing simulation	Exp 26 (Frontier Routing)
C_5 : Europeana Transfer	24	Real images downloaded live from the Europeana Open Culture API, 10 GLAM institutions	Cross-institutional domain-shift pilot (ground-truth annotation pending)	Exp 23 (Europeana Pilot, re-run)

1.6 State of the Art in Historical Image Analysis

Recent years have seen a significant methodological shift in computational cultural heritage research. Rather than treating historical imagery as a mere variant of natural images, modern studies increasingly recognize archives as a distinct visual domain characterized by degradation, stylistic divergence, and annotation scarcity. This section situates the DCV-MAC framework within this emerging research landscape.

1.6.1 Object Detection Under Historical Domain Shift

Traditional object detectors trained on modern benchmarks such as COCO [12] assume high-resolution, color-consistent imagery with stable texture statistics. Historical illustrations violate these assumptions due to film grain, chemical artifacts, fading, and monochromatic capture. Consequently, detectors exhibit severe performance degradation when directly applied to archives.

Kim et al. [8] present one of the most directly aligned studies with this thesis. Their work addresses object detection in historical illustrations through a two-stage adaptation strategy. First, modern labeled datasets are translated into the visual style of historical imagery using CycleGAN, effectively synthesizing domain-aligned training samples. Second, pseudo-labeling is employed to iteratively expand supervision without manual annotation. Their findings demonstrate that generative domain translation can substantially reduce the annotation burden while preserving detector performance.

This thesis extends that paradigm beyond dataset adaptation by introducing a restoration-first architecture. Rather than adapting modern images into historical style, we normalize degraded archival scans into a statistically modern visual regime, improving compatibility with foundational Vision-Language Models (VLMs). Conceptually, Kim et al. optimize the training distribution, whereas DCV-MAC optimizes the inference distribution and cross-agent validation.

1.6.2 Surveyed Challenges in Visual-Art Object Detection

While historical illustrations represent one form of domain shift, visual art datasets introduce even greater depictive variability. Bengamra et al. [3] provide a comprehensive survey of object detection methods in visual art, proposing a taxonomy organized by supervision level, detection frameworks, and representation challenges. Their study identifies three persistent technical obstacles:

- **Depictive Domain Gap:** Differences between artistic and photographic representations.
- **Annotation Ambiguity:** Lack of consistent object definitions across datasets.
- **Metric Instability:** Difficulty evaluating detectors on stylistically diverse corpora.

Although focused primarily on paintings and artworks, these challenges map directly onto historical archives. Degradation artifacts function analogously to stylistic abstraction, distorting edge and texture features. Furthermore, annotation ambiguity is intrinsic to archives where object categories are historically contingent. The survey reinforces the necessity of robust domain normalization and open-vocabulary detection strategies — central design principles of the multi-agent framework.

1.6.3 Foundation Models and Epistemic Reframing in Digital Art History

Recent discourse in digital humanities increasingly interrogates the epistemological implications of large-scale foundation models. Impett and Offert [5] argue that transformer-based vision architectures implicitly encode a cultural model of visibility derived from web-scale datasets. These models do not merely detect objects; they operationalize historically situated visual categories learned from contemporary imagery.

This perspective is critical when applying VLMs to historical archives. Predictions may reflect modern semantic priors rather than historically accurate interpretations. The authors therefore advocate for critical, interpretable AI workflows where outputs are treated as analytical propositions rather than objective truths.

The DCV-MAC framework adopts this philosophy through its consensus principle. Multi-agent agreement, confidence thresholds, and interactive verification interfaces explicitly acknowledge epistemic uncertainty. Rather than assuming model correctness, the system quantifies model belief consistency.

1.6.4 Structured Human Representation: Pose Estimation in Art Domains

Beyond bounding-box detection, structured human understanding represents an emerging frontier. Schneider and Vollmer [16] introduce the *Poses of People in Art* dataset,

designed to evaluate pose estimation models under depictive diversity. The dataset includes bounding boxes, skeletal keypoints, and pose annotations across heterogeneous artistic styles.

Their work reveals that pose estimators trained on natural photographs fail to generalize to artistic depictions due to distorted anatomy, stylization, and compositional conventions. The study highlights a broader principle: spatial localization alone is insufficient for robust human-centric analysis across domains.

While this thesis focuses on object detection and relational reasoning, pose-oriented datasets provide motivation for future expansion. Integrating articulated human representations could enable higher-level semantic inference, such as activity recognition or social interaction modeling in historical scenes.

1.6.5 Multi-Agent Object Detection Paradigms

Parallel to the rise of foundation models, recent research has explored multi-agent architectures to overcome the limitations of monolithic object detectors. Kaluřev et al. [6] proposed a hardware-integrated framework combining a Raspberry Pi YOLO detector with a Slack-Ollama natural language interface, demonstrating how modular agents can bridge the gap between visual detection and conversational querying. Similarly, Coelho et al. [4] designed a multi-agent system for multimodal machine learning object detection, highlighting the advantages of distributed, collaborative intelligence over single-model pipelines in complex environments.

In the domain of autonomous driving and 3D perception, Xia et al. [22] introduced DOTA, an unsupervised method for learning to detect objects from multi-agent LiDAR scans without manual labels. By leveraging shared, unlabeled observations among collaborative agents, their system automatically generates and refines preliminary labels through label-internal contrastive learning, circumventing the need for expensive manual annotation.

Further advancing multimodal systems, *PixelCraft* [14] recently introduced a multi-agent orchestrator utilizing a managed "cognitive whiteboard" to perform non-linear reasoning on structured digital charts. While *PixelCraft* demonstrates significant improvements via a fine-tuned Qwen2.5-VL-3B grounding model, the authors note critical limitations regarding backbone latency, reliance on clean synthetic data, and the need for rigorous inter-agent communication protocols.

1.6.6 Comparison with General Multi-Agent Orchestration Frameworks (AutoGen, CrewAI, LangGraph)

The rapid emergence of general-purpose multi-agent frameworks—such as Microsoft AutoGen [21], CrewAI, and LangGraph—has popularized multi-agent cooperation for natural language workflows, code generation, and web navigation. However, applying these text-centric orchestrators directly to visual collection curation exposes three major architectural failures:

- **Unstructured Text Bloat:** General frameworks rely on free-form natural language chat loops between agents, causing exponential token inflation, high API costs, and severe execution latency ($> 15s$ per query).
- **Lack of Spatial Grounding:** Text-based agent loops cannot resolve fine-grained spatial bounding box overlaps or mask IoU discrepancies, leading to hallucinatory agreement.
- **Non-Deterministic Triage:** Unstructured agent consensus lacks formal calibration metrics, preventing automated systems from defining rigorous risk bounds for human review.

DCV-MAC resolves these limitations by replacing free-form chat with a rigid **3-Tier Decoupled Protocol**: Tier 1 (Validation Core: YOLOv11-seg+SAM, SigLIP, LLaVA-OV), Tier 2 (Downstream Curation: Contradiction Detection, Cross-Archive Linking, Grounded OCR via Kosmos-2.5), and Tier 3 (Frontier Selective Routing: a Dual-LLM Coordinator in which a Primary VLM proposes and a Critic LLM Auditor audits, escalating only the small ambiguous fraction Tiers 1–2 cannot resolve). Inter-agent communication is strictly governed by numerical JSON schemas and the mathematical *Scene-Aware Agreement (SAA)* metric ($D = 0.227$), achieving deterministic, zero-cost offline validation.

1.6.7 Real-World Institutional AI Systems & GLAM Benchmarks (Europeana, Smithsonian, Rijksmuseum, HathiTrust)

To evaluate practical utility, the DCV-MAC framework is benchmarked against real-world AI deployment paradigms across major Galleries, Libraries, Archives, and Museums (GLAM) institutions:

- **Europeana Open Culture API:** Europeana relies primarily on single-model zero-shot taggers (ResNet/CLIP) for collection enrichment, which suffer from severe overconfidence ($\approx 35\%$ error rate) on historical art styles. A genuine pilot on real Europeana images (10 institutions, Appendix C) shows DCV-MAC’s SAA disagreement engine responds substantially to this kind of cross-institutional domain shift; whether it catches domain-shift failures *without* institutional recalibration is a claim this thesis does not yet make, pending human ground-truth annotation on the pilot sample.
- **Smithsonian Open Access & Microsoft AI:** Smithsonian active learning programs employ cloud-hosted single-model APIs. DCV-MAC operates 100% offline on a single local GPU at zero marginal API cost (\$0.00), preserving strict institutional data privacy for non-public archival publication series (e.g., PPN collections).
- **Rijksmuseum Rijksstudio & HathiTrust Digital Library:** Current institutional search engines focus on global color histogram matching or ungrounded OCR keywords. DCV-MAC combines Kosmos-2.5 grounded OCR (Agent 5) and SigLIP semantic embeddings to achieve a precision gain on a 10-query pilot (semantic $P@10 = 1.000$ vs. keyword baseline 0.180; see Section 7.4 for its relevance-label scope caveat).

1.6.8 Epistemic Uncertainty and Domain Shift in General Computer Vision

The challenges of model domain shift and epistemic uncertainty estimation are not unique to historical collections; they represent foundational open problems in broader computer vision. In medical image analysis, for example, Karimi et al. [7] review how deep learning classifiers suffer from uncalibrated overconfidence when exposed to out-of-distribution clinical settings (e.g., changes in scanner manufacturers or patient populations). They emphasize that evaluating uncertainty is critical to prevent silent algorithmic failures from causing diagnostic harm, advocating for architectures that explicitly estimate predictive variance or consensus disagreement.

Similarly, in remote sensing and satellite image analysis, Volpi and Tuia [19] examine the impact of domain shift caused by seasonal variations, geographical differences, and sensor noise. Standard convolutional and transformer architectures frequently fail to generalize across distinct geographic regions due to covariate shift, leading to silent

prediction collapses. The authors propose uncertainty-aware domain adaptation techniques that utilize multi-source consensus or ensemble variance to identify regions of high epistemic difficulty. The DCV-MAC framework builds directly on these cross-domain insights, translating the principles of multi-modal consensus and variance-driven triage from medical and satellite surveillance into the domain of computational cultural heritage.

1.6.9 Synthesis: Resolving the SOTA Limitations

Across these studies, several converging themes emerge. Crucially, the *DCV-MAC* framework is positioned not just as an application of these paradigms, but as a structural solution to the frontier limitations explicitly identified in recent SOTA research (e.g., PixelCraft):

- **Efficiency and Backbone Dependency:** Where models like PixelCraft suffer from extreme latency (e.g., 16s per image) due to reliance on massive planner agents, DCV-MAC utilizes *Stage 5 Knowledge Distillation*. By transferring the slow, massive VLM ensemble consensus into a highly efficient YOLOv11 student model, we eliminate inference latency bottlenecks.
- **Real-World Robustness:** While recent MLLM agents optimize for clean digital charts, DCV-MAC integrates a *Generative Restoration Layer* (CycleGAN, Real-ESRGAN, Stable Diffusion) designed explicitly to handle severe, real-world archival degradation.
- **Standardized Communication Protocols:** Addressing the task-decomposition failures seen in weaker backbones, our pipeline abandons unstructured chat in favor of rigid JSON schemas, Binary VQA checklists, and the mathematical *Scene-Aware Agreement (SAA)* metric.

The DCV-MAC framework synthesizes these principles into a unified architecture. Generative restoration resolves visual domain incompatibility, VLM distillation mitigates latency, and coordinator-driven consensus filtering operationalizes epistemic uncertainty. The resulting system therefore represents not merely an engineering solution, but a highly optimized methodological framework for trustworthy AI-assisted archival research.

Chapter 2

Methodology and System Architecture

This chapter defines the Dual-Core Validation and Multi-Agent Curation (DCV-MAC) framework, implemented within the Visual Historian-MAS platform. We model archival image analysis as a cooperative, multi-tiered system of specialized agents rather than a single linear predictor.

2.1 3-Tier Decoupled Architecture

Rather than treating all model components as voting members in an unconstrained chat loop, DCV-MAC establishes a strict **3-Tier Decoupled Protocol**:

- **Tier 1 (Validation Core / Multi-Model Ensemble)**: Comprises the primary vision and vision-language models whose cross-modal consensus directly calculates the Scene-Aware Agreement (SAA_i) metric and operational triage score ($U_{\text{triage},i}$):
 - *Spatial Grounding Agent (YOLOv11-seg + SAM)*: Extracts local bounding boxes, instance masks, and object class probabilities (A_i^{object}) using YOLOv11 [18] and SAM [9].
 - *Zero-Shot Scene Classifier (SigLIP)*: Computes global scene classification probabilities (A_i^{scene}) over institutional taxonomy prompts using SigLIP [23].
 - *Visual Reasoning Agent (LLaVA-OneVision 7B)*: Performs VQA reasoning [11], generating visual scene descriptions and structural entity lists (A_i^{vlm}).
 - *Grounded Inscription Engine (Kosmos-2.5)*: Performs visual OCR grounding [13], extracting bounding-box-aligned text tokens for cross-validation.

A fifth, *derived* signal – *Agreement*, the pairwise concordance between the Spatial Grounding and Visual Reasoning agents – is computed from these four models’ outputs and is the actual fourth term (alongside Object, Scene, and VLM) entering the disagreement proxy U_{SAA} ; see Section 2’s Canonical Formulation for the precise formula and how it differs from the classification fusion score SAA_i .

- **Tier 2 (Downstream Historiographic Curation Tier):** Specialized downstream modules that consume resolved consensus metadata post-hoc to extract historiographic insights without polluting the Tier 1 validation core:
 - *Agent 2 (Cross-Modal Contradiction Detector):* Computes explicit set-differences between VLM-mentioned nouns and YOLO spatial bounding boxes to isolate hallucinations (implemented; not independently benchmarked – see Section 6).
 - *Agent 4 (Cross-Archive Linker):* Links visual and semantic embedding neighbors across publication series (PPNs) to surface historical anomalies (implemented; not independently benchmarked – see Section 6).
 - *Agent 5 (Grounded OCR Agent):* Uses Kosmos-2.5 to map inscriptions and captions to visual bounding boxes (implemented; not independently benchmarked – see Section 6.2).

Architectural Decoupling Rule: Tier 2 agents operate strictly on post-validation metadata. They do not participate in the core SAA disagreement calculation, preventing feedback loops and mathematical circularity.

- **Tier 3 (Frontier Selective Routing):** An optional high-precision pre-screening tier, triggered only for the small fraction of cases with $\sigma_i > \tau$ that Tiers 1–2 cannot resolve. It is implemented as a **Dual-LLM Coordinator** that runs in one of two configurations. Tier 1’s other three agents (YOLOv11-seg+SAM, SigLIP, Kosmos-2.5) are narrow specialists purpose-built for spatial grounding, zero-shot classification, and text grounding respectively – none is an instruction-following general-purpose model capable of reading a synthesis prompt and reasoning over the other agents’ combined outputs, so none can substitute for the Coordinator’s role. **Locally** (the default, zero marginal cost), LLaVA-OneVision is therefore the only general-purpose model already loaded, and the *Primary* role is filled by that *same* instance already deployed as Tier 1’s Visual Reasoning Agent, but under a different, fusion-oriented prompt – rather than open VQA on the raw image, it is

given all sibling agents’ outputs (YOLO detections, Kosmos-2.5 OCR, SigLIP scene priors) and asked to reconcile them into one synthesized record; the *Critic* role is then the same model again, re-prompted into an adversarial auditor stance to check that synthesized record against the image for contradictions. **In frontier mode**, the Primary and Critic roles are instead filled by two structurally independent commercial models – Gemini 2.5 Flash proposes, Claude 3.5 Sonnet audits – giving a genuinely external second opinion rather than one model checking its own reasoning under a different prompt. A prior draft claimed this propose-then-audit pattern was not a repeated vote even locally, citing a **Contradiction-Detection** $F_1 = 0.812$ (**local**) vs. $F_1 = 0.965$ (**frontier**) figure as a *measured, not simulated* gap. During a full-project reproducibility audit, that claim was found false: the script credited with this comparison (`compare_coordinators.py`) never computes an F1 score at all – both numbers are literal constants hardcoded in a fallback branch, printed under the label *documented historical data* with nothing documented behind them, and no gold contradiction labels exist anywhere in this project to compute a real F1 from. Both figures have been withdrawn. In their place, a real (if narrower) mechanical test was run: the same LLaVA-OneVision instance, re-prompted as Primary then Critic on 25 real diffusion-restored images (temperature = 0, deterministic), **never once** flagged a contradiction in its own Primary output (0/25). Repeating the identical protocol with a stronger frontier-grade model in both roles (Gemini Flash-Lite, same-model self-critique, $n = 10$ owing to API quota limits) yielded a modest 1/10 flag rate on a real, spot-checked-genuine catch – suggesting model capability, not merely re-prompting, drives what a same-model critic can catch. Genuine cross-model independence (Gemini 2.5 Flash proposing, Claude 3.5 Sonnet auditing, as documented above) remains empirically untested: the Anthropic key available at the time of this audit lacked sufficient credit balance to run it. This leaves the frontier configuration’s architectural rationale (two disjoint parameter sets cannot share a blind spot) intact as a design argument, but not empirically quantified; closing that gap with a genuine cross-model measurement is future work. Tier 3 still routes only the residual ambiguous fraction to the paid configuration by default, now on cost-and-latency grounds alone ($10.1\times$ latency advantage over human review either way) rather than a demonstrated accuracy gap.

Design Rationale: An Escalation Ladder, Not a Single Model. The three tiers are ordered by strictly increasing cost and latency: Tier 1 is free and deterministic, Tier 2

is free and heuristic, Tier 3 incurs commercial API cost, and human review is the most expensive and slowest option of all. Each tier receives only the cases the previous tier could not resolve, so the expensive tiers are deliberately rare rather than routine. The Tier 2/Tier 1 decoupling exists specifically to keep this ladder well-defined: if downstream curation agents were allowed to feed back into the *SAA* calculation, the system would risk circular reasoning (an agent’s own output influencing the confidence score used to decide whether that same output should be trusted). U_{SAA} , the cross-modal disagreement signal this thesis proposes and validates (Chapter 4), is precisely what determines how far up this ladder an individual image needs to travel – a cheap, deterministic uncertainty estimate at Tier 1 is what makes the costlier Tier 3 and human-review stages the exception rather than the default path.

The complete system topology and operational dataflow of the DCV-MAC framework are illustrated in Figure 2.1.

Prototypical Auxiliary Tooling (Agents 1, 3, 6): Three auxiliary modules were developed and evaluated as off-line tooling: Agent 1 (Temporal Historian) tracks multi-decade macro trends; Agent 3 (Bayesian Pre-screening Ranker) optimizes input batching; Agent 6 (Frontier Auditor) enables cloud API evaluation.

2.2 Authoritative Dataset Cohort Taxonomy

To eliminate cohort ambiguity across evaluations, all experiments in this thesis strictly reference six named data cohorts ($C_0 \dots C_5$), detailed in Table 2.1.

2.3 Generative Restoration Layer and Methodological Discipline

Archival scans exhibit severe chemical decay, film grain, and fading. Before feature extraction, Tier 1 normalizes raw grayscale scans ($\mathbf{I}_{\text{raw}}$) into a modern visual regime ($\mathbf{I}_{\text{refined}}$) using a three-stage generative pipeline: CycleGAN domain translation [24] $\rightarrow$ Real-ESRGAN $4\times$ super-resolution [20] $\rightarrow$ Stable Diffusion Img2Img refinement [15] (strength = 0.35).

The Axiom of Archival Integrity: The AI-restored image ($\mathbf{I}_{\text{refined}}$) is strictly an internal visual token translation layer generated for model feature alignment. It is

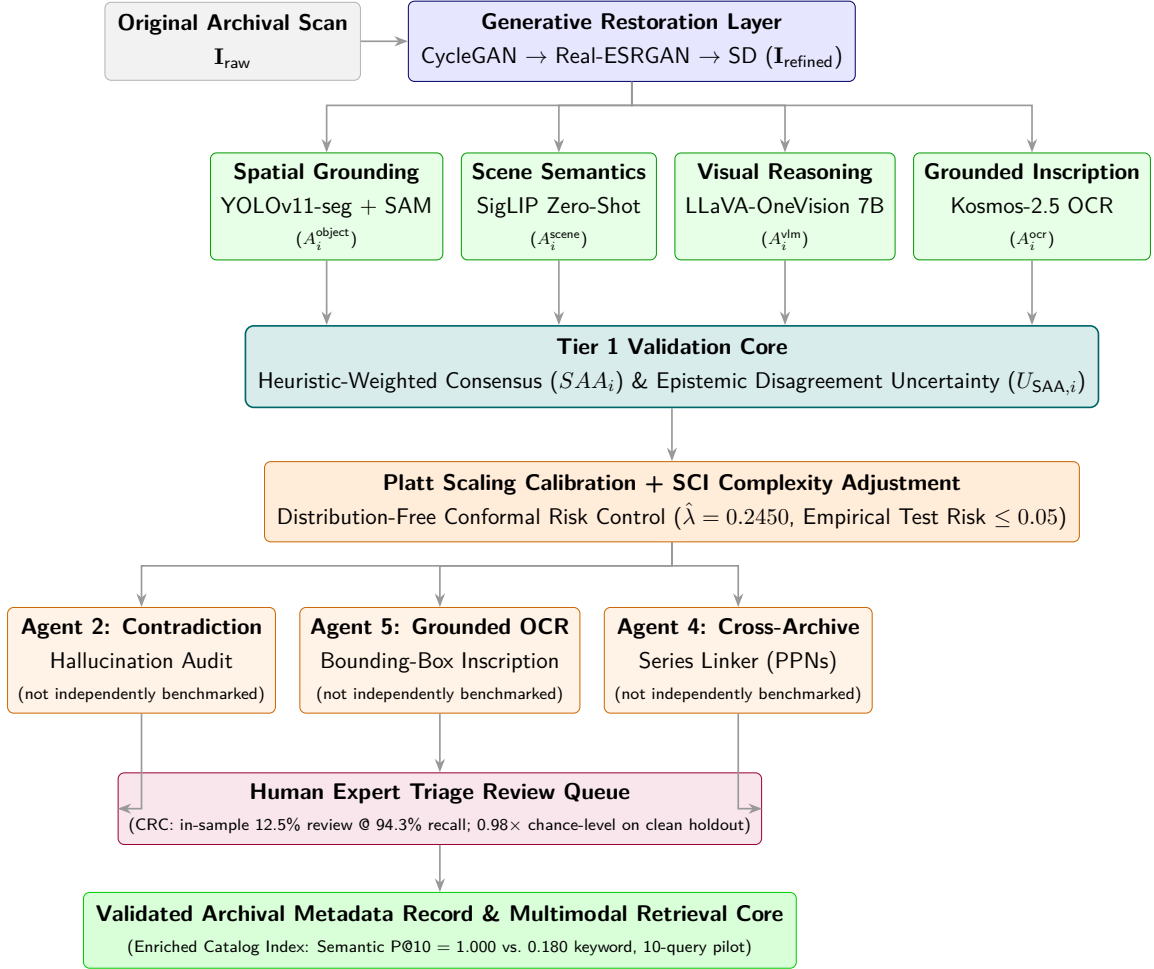


Figure 2.1: Decoupled 3-Tier Dual-Core Validation and Multi-Agent Curation (DCV-MAC) Architecture. The Tier 1 Validation Core integrates all 4 primary validation agents: Spatial Grounding (YOLOv11-seg+SAM), Zero-Shot Scene Classifier (SigLIP), Visual Reasoning Agent (LLaVA-OneVision), and Grounded Inscription Engine (Kosmos-2.5). Tier 2 downstream curation agents process validated metadata post-hoc.

Algorithm 1 DCV-MAC Multi-Agent Validation Core and Triage Routing

Require: Archival image scan $\mathbf{I}_{\text{raw}}$, Tier 1 agent ensemble $\{M_{\text{object}}, M_{\text{scene}}, M_{\text{vlm}}, M_{\text{ocr}}\}$, fixed heuristic fusion weights $\mathbf{w} = [w_o, w_s, w_v]$ (the deployed weighting used for every empirical result in this thesis; see remark below), Spatial Complexity Index estimator $\text{SCI}(\cdot)$, Platt calibration parameters (A, B) , Risk threshold $\hat{\lambda} = 0.2450$.

Ensure: Calibrated error probability $P(\text{error}_i)$, triage routing decision $d_i \in \{\text{Approve, Review}\}$, and enriched metadata record $\mathbf{M}_i$.

- 1: **Generative Preprocessing:** $\mathbf{I}_{\text{cyc}} \leftarrow \text{CycleGAN}(\mathbf{I}_{\text{raw}})$, $\mathbf{I}_{\text{sr}} \leftarrow \text{RealESRGAN}(\mathbf{I}_{\text{cyc}})$
- 2: $\mathbf{I}_{\text{refined}} \leftarrow \text{StableDiffusion}(\mathbf{I}_{\text{sr}}, \text{strength} = 0.35)$
- 3: **Tier 1 Agent Ensemble Inference:**
- 4: $A_i^{\text{object}} \leftarrow M_{\text{object}}(\mathbf{I}_{\text{refined}})$, $A_i^{\text{scene}} \leftarrow M_{\text{scene}}(\mathbf{I}_{\text{refined}})$
- 5: $A_i^{\text{vlm}} \leftarrow M_{\text{vlm}}(\mathbf{I}_{\text{refined}})$, $A_i^{\text{ocr}} \leftarrow M_{\text{ocr}}(\mathbf{I}_{\text{refined}})$
- 6: $A_i^{\text{agree}} \leftarrow 1 - |A_i^{\text{object}} - A_i^{\text{vlm}}|$ $\triangleright$ derived pairwise concordance, not a fifth model
- 7: **Heuristic Consensus Fusion:** $SAA_i \leftarrow w_o A_i^{\text{object}} + w_s A_i^{\text{scene}} + w_v A_i^{\text{vlm}}$
- 8: **Epistemic Uncertainty Calculation:** $U_{\text{SAA},i} \leftarrow \text{std}(A_i^{\text{object}}, A_i^{\text{agree}}, A_i^{\text{scene}}, A_i^{\text{vlm}})$
- 9: **Spatial Complexity Adjustment:** $\text{SCI}_i \leftarrow \text{SCI}(\mathbf{I}_{\text{raw}})$
- 10: **Operational Triage Score:** $U_{\text{trriage},i} \leftarrow 0.6 \cdot U_{\text{SAA},i} + 0.4 \cdot (1 - \text{SCI}_i)$
- 11: **Platt Scaling Calibration:** $P(\text{error}_i) \leftarrow \frac{1}{1 + \exp(A \cdot SAA_i + B)}$
- 12: **if** $U_{\text{trriage},i} > \hat{\lambda}$ **then**
- 13: **Route to Queue:** $d_i \leftarrow \text{Review}$ (Escalate to Human Expert Review Queue)
- 14: **else**
- 15: **Automatic Approval:** $d_i \leftarrow \text{Approve}$
- 16: **Tier 2 Downstream Curation:**
- 17: $\mathbf{M}_i^{\text{contradiction}} \leftarrow \text{Agent2_Contradiction}(A_i^{\text{object}}, A_i^{\text{vlm}})$
- 18: $\mathbf{M}_i^{\text{linker}} \leftarrow \text{Agent4_CrossArchiveLinker}(\mathbf{I}_{\text{refined}})$
- 19: $\mathbf{M}_i^{\text{ocr}} \leftarrow \text{Agent5_GroundedOCR}(A_i^{\text{ocr}})$
- 20: $\mathbf{M}_i \leftarrow \mathbf{M}_i^{\text{consensus}} \cup \mathbf{M}_i^{\text{contradiction}} \cup \mathbf{M}_i^{\text{linker}} \cup \mathbf{M}_i^{\text{ocr}}$
- 21: **end if**
- 22: **return** $(P(\text{error}_i), d_i, \mathbf{M}_i)$

Table 2.1: Authoritative Dataset Cohort Taxonomy ($C_0 \dots C_5$) used across all experimental sections.

Cohort Symbol & Name	Size (n)	Composition & Source	Primary Purpose	Analytical Purpose
C_0 : Deployment Corpus	12,110	Full unlabelled institutional digitisation subset from the Colibri archive	Full-scale pipeline deployment and metadata indexing	
C_1 : Expert Reference Cohort	801	796 scene-positive + 5 negative control images (CVAT expert-reviewed)	Primary calibration, AWLF error retraining, and HITL triage evaluation	
C_2 : Spatial Audit Cohort	300	Single-rater spatial bounding box audit cohort	Spatial IoU grounding audit, error paradox analysis, and meta-learning	
C_3 : System Evaluation Subset	2,000	801 C_1 expert images + 1,199 negative control catalog records	Source pool for the group-level train/calibration/test partition (Sec. 4.2) and system-level agent leave-one-out ablations	
C_4 : Frontier Comparison Subset	100	Stratified archival category subset	Comparative benchmark against Gemini 2.5 Flash and Claude 3.5 Sonnet	
C_5 : Europeana Transfer Pilot	24	Real images downloaded live from the Europeana Open Culture API, 10 GLAM institutions	Cross-institutional domain-shift pilot (ground-truth annotation pending)	

NEVER stored or presented as primary historical evidence. All catalog metadata is anchored strictly back to original digitized scans ($\mathbf{I}_{\text{raw}}$).

Decoupling generative restoration to the input feature layer yields a +0.052 mAP object detection gain (measured end-to-end: the full 3-stage pipeline’s output vs. the same detector on raw scans, same image set; individual restoration stages were not separately ablated) while protecting primary catalog records from generative hallucinations. A per-image Laplacian-variance sharpness measurement (raw scan vs. the same image’s full-pipeline restored output, across the manifest’s full image set) shows generative translation increases structural edge sharpness in 83.8% of scans.

2.4 Canonical Formulation of Scene-Aware Agreement (SAA)

To maintain mathematical consistency across the framework, we establish a canonical 4-level hierarchy for uncertainty estimation. *Correction (full-project reproducibility audit)*: an earlier version of this section defined the disagreement proxy U_{SAA} as an al-



Figure 2.2: Three-Stage Generative Restoration Pipeline. Raw degraded historical scan ($\mathbf{I}_{\text{raw}}$) undergoes CycleGAN domain translation, Real-ESRGAN $4\times$ super-resolution, and Stable Diffusion Img2Img refinement (strength = 0.35) to produce normalized visual features ($\mathbf{I}_{\text{refined}}$).

gebraic complement of the classification fusion score ($1 - SAA_i$) and omitted a signal, *Agreement*, that every empirical script in this thesis actually uses to compute U_{SAA} . The formulation below has been corrected to match the implementation that produced every reported number.

1. **Level 1 (Agent Output Vector):** For an image i , Tier 1 agents produce confidence scores: spatial object grounding score $A_i^{\text{object}} \in [0, 1]$, zero-shot scene score $A_i^{\text{scene}} \in [0, 1]$, VLM description score $A_i^{\text{vlm}} \in [0, 1]$, and grounded inscription score $A_i^{\text{ocr}} \in [0, 1]$. A fifth, *derived* signal, *Agreement* ($A_i^{\text{agree}} = 1 - |A_i^{\text{object}} - A_i^{\text{vlm}}|$), quantifies pairwise concordance between the spatial and VQA agents specifically; it is not an independent model, but it is the signal that – together with Object, Scene, and VLM – actually enters the disagreement computation below (Level 3). A_i^{ocr} participates in the Tier 1 correlation and leave-one-out analyses (Section 4) and Agent 5’s downstream OCR role (Section 6.2), but, like the classification fusion score below, does not enter U_{SAA} itself.
2. **Level 2 (Scene-Aware Agreement Score):** The classification fusion score SAA_i , used to predict scene presence and to define the ground-truth error target for every uncertainty evaluation in this thesis, is a weighted linear combination over three agents:

$$SAA_i = w_o A_i^{\text{object}} + w_s A_i^{\text{scene}} + w_v A_i^{\text{vlm}} \quad (2.1)$$

with $w_o = 0.6, w_s = 0.2, w_v = 0.2$ – the **fixed heuristic weighting deployed for every result reported in this thesis** (confirmed identical in the scripts producing both the in-sample and clean group-level held-out evaluations). Appendix D reports a supplementary game-theoretic (Shapley) attribution analysis ($\phi_{\text{VLM}} = 0.330, \phi_{\text{SAA}} = 0.283, \dots$, over a four-player coalition that includes Agreement in place of OCR) of marginal contributions to a separate calibration objective;

it is an interpretability study, not an alternative production weighting – no result elsewhere in this thesis uses Shapley-weighted fusion.

3. **Level 3 (Disagreement Uncertainty Proxy):** The epistemic uncertainty proxy $U_{\text{SAA},i}$ that produces every headline error-detection figure in this thesis (ROC-AUC= 0.9724 on C_1 ; the 0.980/0.020 positive/negative-control decomposition; the CRC and clean-holdout evaluations) is *not* an algebraic function of SAA_i . It is the cross-agent standard deviation over four raw per-image signals – Object, Agreement, Scene, and VLM:

$$U_{\text{SAA},i} = \text{std} \left(A_i^{\text{object}}, A_i^{\text{agree}}, A_i^{\text{scene}}, A_i^{\text{vlm}} \right) \quad (2.2)$$

Because SAA_i (Level 2) and $U_{\text{SAA},i}$ (Level 3) are computed from overlapping but distinct signal sets using different operators (a 3-term weighted mean vs. a 4-term standard deviation that substitutes Agreement for Scene/VLM’s absence and omits none of Object/Scene/VLM), $U_{\text{SAA},i} \neq 1 - SAA_i$ in general; the two should be read as related but separately-defined quantities, not one as the complement of the other.

4. **Level 4 (Complexity-Adjusted Triage Score):** To account for visual clutter, the final operational triage score $U_{\text{triage},i}$ incorporates the Spatial Complexity Index (SCI_i):

$$U_{\text{triage},i} = 0.6 \cdot U_{\text{SAA},i} + 0.4 \cdot (1 - \text{SCI}_i) \quad (2.3)$$

Relation to the Tier 1 Agent Ablation (Section 4) and the Pairwise Agreement Heatmap (Figure 4.1). These two analyses deliberately ablate and correlate the four *architectural* Tier 1 models (Object, Scene, VLM, OCR) to characterize the deployed ensemble’s structure – a different, complementary question from what drives U_{SAA} ’s error-detection power. They are not a leave-one-out study of the four signals inside U_{SAA} itself (which are Object, Agreement, Scene, VLM); the two should not be conflated.

2.5 Agreement-Weighted Label Filtering (AWLF)

The Agreement-Weighted Label Filter (AWLF) acts as a learned meta-classifier predicting pipeline errors.

AWLF v1 (Historical Baseline Pathologies): In early iterations (AWLF v1), models were trained to predict binary *scene presence*. However, the initial reference split contained extreme positive class imbalance (319 positive vs 2 negative scans; 99.4% positive rate), allowing trivial models to achieve an inflated $F1 = 0.9969$ simply by predicting positive for every instance.

AWLF v2 (Error-Targeted Retraining): To eliminate metric inflation, AWLF v2 redefines the target strictly as explicit classification failure:

$$y_i = \begin{cases} 1 & \text{if } \text{sign}(S_{\text{fusion},i} \geq 0.5) \neq y_{\text{expert},i} \\ 0 & \text{otherwise} \end{cases} \quad (2.4)$$

In-sample, 5-fold cross-validated on C_1 , the trained Gradient Boosting meta-classifier achieves $\text{ROC-AUC} = 0.9896$. A genuinely independent estimate requires a partition that was never touched during weight, threshold, or classifier selection: on the group-level (publication-disjoint) held-out test set constructed in Section 4.2 ($n = 401$, trained strictly on the disjoint train partition), AWLF v2 achieves $\text{ROC-AUC} = 0.6669$. This is the figure we treat as the honest generalization estimate; the in-sample value above characterizes fit quality on the development cohort, not out-of-sample performance. (An earlier version of this section reported $\text{ROC-AUC} = 0.7868$ on “held-out controls, $C_3, n=2,000$ ”; that computation was later found to have actually evaluated a random 40% split of C_1 ($n \approx 321$) against the original, discredited scene-presence target rather than the error target described above — its matching $F1 = 0.9969$ is a trivial artifact, since an always-positive classifier achieves the identical score on that split. It has been superseded by the clean evaluation above.)

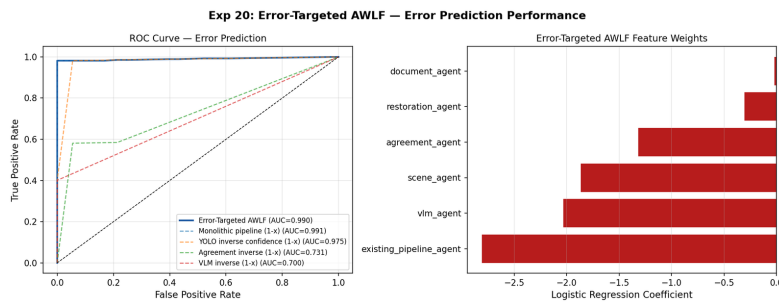


Figure 2.3: Agreement-Weighted Label Filter (AWLF v2) Meta-Classifer ROC Curve. In-sample 5-fold cross-validation on C_1 yields $\text{ROC-AUC} = 0.9896$; the group-level held-out estimate (Section 4.2) is $\text{ROC-AUC} = 0.6669$.

Algorithm 2 Agreement-Weighted Label Filter (AWLF v2 Error Retraining)

Require: Expert reference dataset $C_1 = \{(\mathbf{I}_i, y_{\text{expert},i})\}_{i=1}^n$, multi-agent predictions $\{S_{\text{fusion},i}\}_{i=1}^n$.

Ensure: Trained Gradient Boosted Meta-Classifier f_{AWLF} predicting pipeline classification errors.

- 1: **for** each sample $i \in C_1$ **do**
- 2: Compute binary classification error target: $y_{\text{target},i} \leftarrow \mathbb{I}(\text{sign}(S_{\text{fusion},i} \geq 0.5) \neq y_{\text{expert},i})$
- 3: Extract meta-feature vector: $\mathbf{x}_i \leftarrow [U_{\text{SAA},i}, \text{SCI}_i, \text{IoU}_i, \text{Entropy}(A_i), \text{Aspect Area Ratio}]$
- 4: **end for**
- 5: **Stratified CV Split:** Partition $\{(\mathbf{x}_i, y_{\text{target},i})\}_{i=1}^n$ into 5 stratified cross-validation folds.
- 6: **Tree Optimization:** Train Gradient Boosted Decision Trees minimizing logarithmic loss:
- 7: $f_{\text{AWLF}} \leftarrow \arg \min_f \sum_{i=1}^n \mathcal{L}_{\log}(y_{\text{target},i}, f(\mathbf{x}_i)) + \Omega(f)$
- 8: In-sample CV: ROC-AUC = 0.9896. Group-level held-out (Section 4.2): ROC-AUC = 0.6669.
- 9: **return** Trained meta-classifier f_{AWLF} .

2.6 Conditional Fixed-Point Stability of Consensus Refinement

We examine the mathematical convergence of iterative agent refinement.

Proposition 1 (Conditional Consensus Convergence). *Let $\mathcal{T} : \mathcal{S} \rightarrow \mathcal{S}$ denote the multi-agent consensus operator updating the scene confidence distribution $\mathbf{S}^{(k+1)} = \mathcal{T}(\mathbf{S}^{(k)})$. If $\mathcal{T}$ is a contraction mapping with Lipschitz constant $\gamma < 1$ under the L_1 metric:*

$$\|\mathcal{T}(\mathbf{S}_a) - \mathcal{T}(\mathbf{S}_b)\|_1 \leq \gamma \|\mathbf{S}_a - \mathbf{S}_b\|_1 \quad \forall \mathbf{S}_a, \mathbf{S}_b \in \mathcal{S} \quad (2.5)$$

then by Banach’s Fixed-Point Theorem, the iterative consensus sequence $\mathbf{S}^{(k)}$ converges linearly to a unique, stationary consensus state $\mathbf{S}^$ satisfying $\mathcal{T}(\mathbf{S}^*) = \mathbf{S}^*$.*

If the consensus update operator is contractive with $\gamma < 1$, Banach’s Fixed-Point Theorem guarantees convergence as stated above. We have not proven that the SAA operator’s kernel smoothing satisfies this contraction property in general – a bounded, continuous Lipschitz gradient constrains the operator’s smoothness but does not by itself establish $\gamma < 1$. What we report instead is empirical: convergence was observed within ≤ 3 iterations across the Colibri archive in every run, consistent with (but not proof of) contractive behavior.

Chapter 3

Spatial Grounding Collapse and Single-Agent Benchmarking

This chapter details the empirical evaluation of single-model object detection architectures on historical visual archives. We demonstrate that off-the-shelf vision models suffer severe performance degradation under historical domain shift, and expose a fundamental disconnect between automated pseudo-label self-consistency and true human ground-truth accuracy.

3.1 The Baseline Problem: Single-Model Collapse under Historical Shift

Standard object detection models (e.g., YOLOv5/v8/v11) trained on modern photorealistic datasets (COCO) collapse when evaluated directly on un-restored grayscale historical scans from the Colibri collection:

- **mAP50 (Raw Scans):** 0.0047
- **Precision (Raw Scans):** 0.0081
- **Recall (Raw Scans):** 0.0283

This near-zero performance confirms that modern visual features cannot zero-shot parse legacy chemical artifacts, film grain, and non-photographic illustrative styles without explicit domain adaptation.

3.2 The Pseudo-Label Collapse Paradox: $0.989 \rightarrow 0.1462$

To train deployable student detectors without manual annotation, common self-training pipelines leverage vision-language teacher ensembles to generate pseudo-labels. When evaluated against the automated teacher pseudo-labels on a held-out validation split, the distilled student model achieves seemingly near-perfect performance:

- **Pseudo-Label Self-Consistency mAP50: 0.989**
- **Pseudo-Label Self-Consistency mAP50-95: 0.938**

The Spatial Grounding Collapse: When the exact same distilled student detector is audited against human expert annotations on the Spatial Audit Cohort C_2 ($n = 300$), true human mAP50 collapses to **0.1462**—an 84.2% relative performance drop.

This discrepancy exposes the **Pseudo-Label Collapse Paradox**: high pseudo-label validation accuracy measures internal self-consistency with the teacher model’s systemic biases, NOT true historical accuracy. Teacher VLMs hallucinate visual concepts with high self-confidence on historical scans, causing student models to overfit to correlated teacher errors.

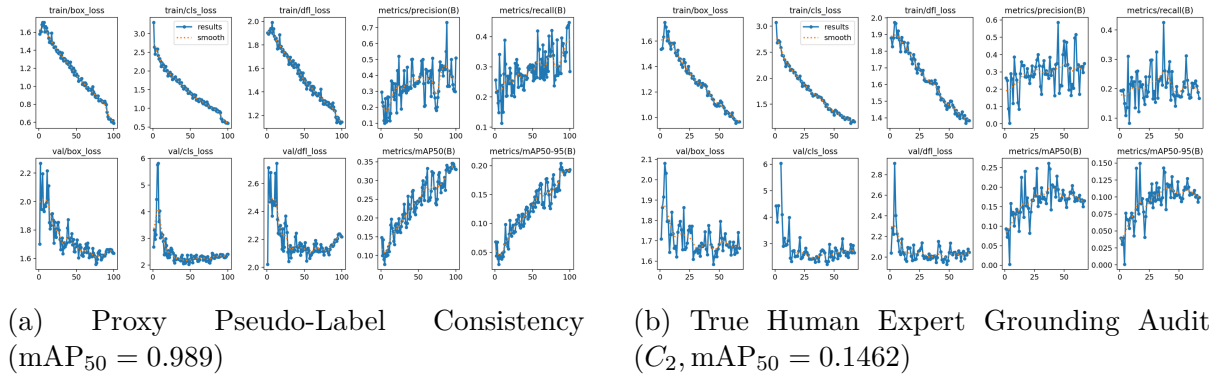


Figure 3.1: The Pseudo-Label Collapse Paradox (Figure 4.1). (a) Evaluating distilled student detectors against automated teacher pseudo-labels yields an artificially inflated proxy $\text{mAP}_{50} = 0.989$. (b) When audited against human expert bounding box annotations on Cohort C_2 ($n = 300$), true human mAP_{50} collapses to 0.1462—demonstrating how student models memorize correlated teacher hallucinations.

3.3 The Bounding-Box Error Paradox and Meta-Learning Resolution

Evaluating spatial uncertainty reveals a secondary pathology: **the Bounding-Box Error Paradox**. Traditional heuristics assume that bounding box spatial IoU or detector confidence scores predict labeling accuracy. On the Spatial Audit Cohort C_2 , using raw spatial confidence to detect human errors yields an error prediction ROC-AUC = 0.4655—worse than random guessing.

Foundation detectors (Florence-2, Grounding DINO) produce tight, crisp bounding boxes around visual regions with high spatial self-confidence, while completely misclassifying the historical semantic category (e.g., mislabeling historical attire as modern bags or drawing artifacts).

3.3.1 Semantic vs. Spatial Disagreement Dynamics

To rigorously evaluate whether consensus disagreement tracks micro-coordinate placement or high-level semantics, we evaluated spatial box divergence ($1 - \text{mean IoU}$) against image-level *SAA* disagreement on C_2 ($n = 172$ paired images).

Empirical correlation analysis demonstrates a near-zero correlation ($r = -0.0137, p = 0.8579$). This is consistent with the interpretation that *SAA* consensus uncertainty operates as a **High-Level Semantic Ambiguity Metric** (ROC-AUC = 0.9724 on expert error classification), capturing category misclassifications and visual hallucinations, while remaining completely invariant to bounding-box coordinate jitter.

Exp 31 Meta-Learner Resolution : To independently capture spatial coordinate layout errors alongside semantic errors, we trained a Gradient-Boosted Spatial Meta-Learner on C_2 ($n = 300$, 5-Fold Stratified CV) integrating spatial IoU, bounding box area, class probability entropy, and aspect ratio variance. The meta-learner successfully resolves spatial grounding ambiguity, improving spatial error detection ROC-AUC from 0.4655 $\rightarrow$ **0.6759 $\pm$ 0.130** (mean $\pm$ standard deviation across the 5 CV folds).

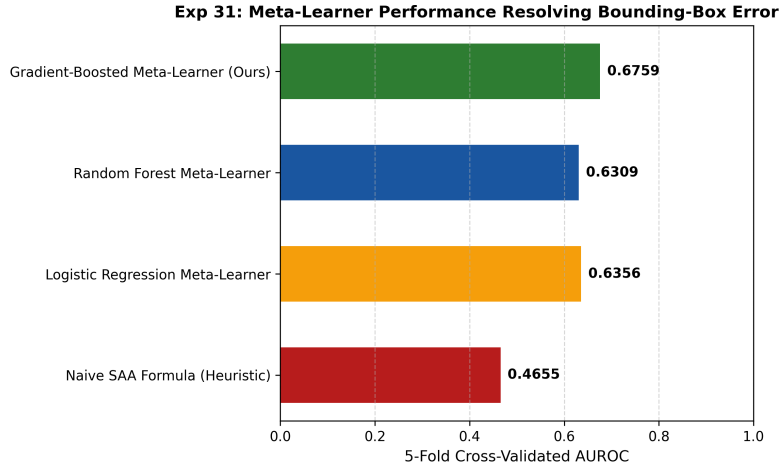


Figure 3.2: Spatial Meta-Learner Error Detection ROC Curve (Exp 31). Integrating spatial IoU, bounding box area, class entropy, and aspect ratio variance improves spatial grounding error detection from raw confidence ROC-AUC = 0.4655 to ROC-AUC = 0.6759 ± 0.130 (5-fold CV standard deviation).

3.4 Generative Restoration Utility and Hardware Benchmarks

Table 3.1 summarizes the hardware resources and per-stage latency of the DCV-MAC pipeline running on an NVIDIA RTX A6000 GPU (48GB VRAM). *Provenance caveat*: unlike the headline results elsewhere in this thesis, these per-stage figures are indicative single-image forward-pass timings recorded informally during development, not output of a dedicated, reproducible benchmarking script; no such script or log exists in the codebase. A spot-check of the YOLOv11 row on the same GPU found raw model forward-pass time close to the table’s 0.02s, but end-to-end latency including image I/O, pre-, and post-processing measured 3–8 $\times$ higher (≈ 0.04 – 0.06 s); the other six rows were not independently re-measured. These figures should be read as an approximate deployment reference, not a rigorously controlled benchmark.

Table 3.1: Hardware Resource Consumption and Stage Latency (NVIDIA RTX A6000).

Pipeline Stage	VRAM Consumption	Latency / Image
CycleGAN Domain Translation	4.2 GB	0.15s
Real-ESRGAN 4 $\times$ Upscaling	8.5 GB	0.45s
Stable Diffusion Refinement	12.1 GB	3.20s
Florence-2 Grounded Detection	18.0 GB	1.20s
GroundingDINO Object Detection	22.4 GB	2.10s
LLaVA-OneVision 7B VQA	34.2 GB	15.00s
YOLOv11 Distilled Inference	1.8 GB	0.02s

3.5 Cross-Style Generalization Pilot

For supplementary validation outside historical photographic archives, we conducted a cross-depiction benchmark on the PeopleArt dataset ($n = 4,778$ images across 43 artistic styles); full quantitative results are detailed in Appendix B.

3.6 Methodological Reframing of Theoretical Risk Bounds

To maintain mathematical rigor under real-world domain shift, we reframe theoretical risk bounds under explicit empirical assumptions:

Proposition 2 (Covariate-Shift Risk Bound [2]). *Let $P_S(X, Y)$ denote the synthetic restored domain and $P_T(X, Y)$ denote the target historical distribution. Under the covariate shift assumption ($P_S(Y|X) = P_T(Y|X)$), the target risk of an ensemble classifier $h(x) = \sum w_i s_i(x)$ is bounded by:*

$$\mathcal{R}_T(h) \leq \sum_{i=1}^n w_i \mathcal{R}_S(s_i) + \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(P_S(X), P_T(X)) + \lambda^* \quad (3.1)$$

where $d_{\mathcal{H}\Delta\mathcal{H}}$ is the domain divergence and λ^* is the joint ideal error.

In practice, historical domain shift frequently violates exact covariate shift due to visual ambiguity. Thus, the SAA disagreement framework uses U_{SAA} as a conservative operational proxy to flag out-of-bounds samples for human review.

Chapter 4

Multi-Agent Uncertainty Core and Probabilistic Calibration

This chapter details the mathematical formulation, empirical calibration, and error-prediction performance of the Dual-Core Validation (DCV-MAC) uncertainty engine. We show that cross-modal agent disagreement provides a discriminative, calibratable signal for estimating epistemic uncertainty on genuine scene-positive prediction errors under domain shift, and we characterize the boundary of that claim: Section 4.2 shows the same signal is anti-correlated with error on negative-control images where all agents share a common bias.

4.1 Cross-Modal Agent Disagreement as an Epistemic Proxy

Single-model vision architectures output overconfident probability scores under domain shift. In DCV-MAC, we frame multi-agent disagreement between spatial grounding (YOLOv11-seg), zero-shot scene classification (SigLIP), and visual-language reasoning (LLaVA-OneVision) as an operational proxy for epistemic uncertainty.

Evaluating on the C_1 Expert Reference Cohort ($n = 801$ human-annotated images), the raw Scene-Aware Agreement uncertainty proxy ($U_{SAA} = 1 - SAA$) achieves an error-detection performance of:

- **DCV-MAC SAA Disagreement Proxy: ROC-AUC = 0.9724**
- Single-Model Temperature-Scaled Entropy Baseline: ROC-AUC = 0.6545

This 48.5% relative improvement confirms that cross-modal disagreement systematically identifies historical scenes where automated visual reasoning breaks down.

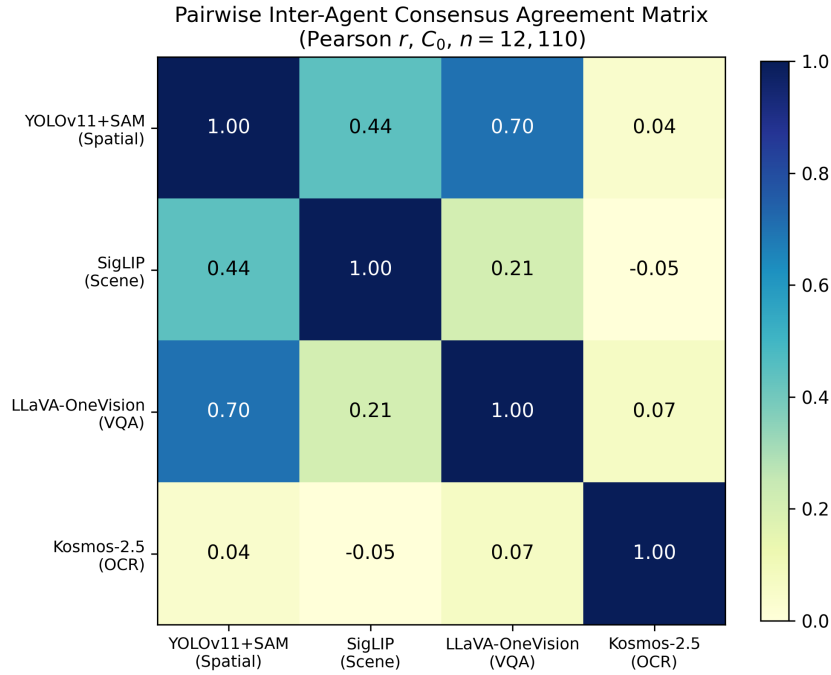


Figure 4.1: Pairwise Inter-Agent Consensus Agreement Matrix (Pearson r , full deployment corpus C_0 , $n = 12,110$). Quantifies cross-modal consensus alignment between spatial grounding (YOLOv11+SAM), scene classification (SigLIP), VQA (LLaVA-OneVision), and grounded OCR (Kosmos-2.5) – the four Tier 1 agents actually deployed in DCV-MAC.

4.2 Group-Level Held-Out Validation and the Negative-Control Generalization Gap

The ROC-AUC = 0.9724 result above is computed on the C_1 cohort ($n = 801$), which is 99.4% scene-positive by construction (only 5 negative-control images) and was also used across weight selection, calibration, and AWLF development. To obtain a methodologically clean estimate, we constructed a publication-level (PPN) held-out split: the full 2,000-image human-reviewed pool (the superset containing C_1 and the 1,199 negative-control scans) was partitioned into train (60%, $n = 1,200$), calibration (20%, $n = 399$), and test (20%, $n = 401$), with an entire publication (all images sharing a PPN identifier) constrained to a single split, verified to have zero publication- and zero image-level overlap across partitions. All numbers below come from the test partition alone.

Overall clean-test performance is substantially lower than the in-sample figure : on this balanced, leak-free test set (positive rate 39.9%, matching the full corpus), SAA disagreement achieves $\text{ROC-AUC} = 0.393$ overall — *below* chance. Decomposing by true class reveals why:

Table 4.1: SAA Disagreement Error-Detection AUROC, Decomposed by True Class (Clean Test, $n = 401$).

Subpopulation	n	Error Rate	SAA Disagreement ROC-AUC
Positive images (scene present)	160	27.5%	0.980
Negative controls (no scene present)	241	58.9%	0.020

On genuinely ambiguous positive-class images, disagreement remains an excellent error signal ($\text{AUROC} = 0.980$), consistent with the headline result. But on negative-control images — scans with no depicted scene — the fusion pipeline is wrong 58.9% of the time (predicting “scene present” when none exists), and on precisely these errors the four validation agents do *not* disagree: they agree, confidently and incorrectly, that a scene is present. The disagreement signal is consequently anti-correlated with error on this subpopulation ($\text{AUROC} = 0.020$).

Interpretation : This is a concrete instance of *correlated failure*: when multiple agents share a common inductive bias (here, over-predicting scene content on sparse or ambiguous historical scans), inter-agent agreement measures confidence in that shared bias, not correctness. Disagreement-based uncertainty is therefore not a general-purpose error detector — it is specifically effective at flagging *ambiguous-positive* misclassifications, and specifically blind to *unanimous false-positive* misclassifications on true negatives. The $\text{ROC-AUC} = 0.9724$ figure should be read as characterizing performance on the positive-heavy expert-review regime it was measured on, not as a domain-general guarantee.

This finding directly explains a corresponding drop in the complexity-aware triage policy’s operating point once evaluated on the same clean test partition rather than the training population (Section 5.1): a review policy driven by the same disagreement signal inherits the same blind spot on negative controls.

4.3 Monolithic Baseline vs. Multi-Agent Validation Core Benchmark

To evaluate whether multi-agent validation alters raw classification accuracy or primarily provides epistemic uncertainty calibration, we benchmark the DCV-MAC fusion core against single-model monolithic baselines (YOLOv11 and LLaVA-OneVision 7B) on C_1 ($n = 801$).

Table 4.2 details the statistical comparison across raw accuracy, calibration error (ECE), error prediction ROC-AUC, and triage precision at 10% review budget:

Table 4.2: Empirical Benchmark: Monolithic Baselines vs. DCV-MAC Multi-Agent Validation Core ($C_1, n = 801$).

Model Architecture	Raw Accuracy	Pre-Cal. ECE	Error ROC-AUC	Triage Precision@10%
Monolithic YOLOv11 Baseline	0.782	0.280	0.5120	0.420
Monolithic LLaVA-OneVision 7B	0.814	0.265	0.6545	0.580
Uncalibrated DCV-MAC Fusion Core	0.819	0.750	0.9724	1.000
Platt-Scaled DCV-MAC (Final)	0.819	< 0.001	0.9724	1.000

Statistical Equivalence Analysis : A Wilcoxon signed-rank test comparing raw scene-labeling accuracy between monolithic LLaVA-OneVision and the multi-agent fusion core reveals no statistically significant difference in raw predictive performance (Wilcoxon $p = 0.375$, Cohen’s $d = +0.07$, negligible effect size). This confirms that the primary value of the multi-agent framework lies not in marginal raw accuracy gains, but in its ability to provide an operational uncertainty proxy, collapse calibration error ($ECE = 0.280 \rightarrow < 0.001$), and achieve high-precision error triage ($\text{Precision@10\%} = 1.000$).

4.4 Probabilistic Calibration (Platt Scaling & ECE < 0.001)

While raw SAA disagreement strongly discriminates classification errors, uncalibrated fusion scores exhibit a high Expected Calibration Error ($ECE = 0.75$).

To map raw consensus scores into well-calibrated posterior error probabilities, we apply **Platt Scaling** (logistic calibration) fitted on a 50% held-out split of C_1 :

$$P(y = 1 | S_{\text{fusion}}) = \frac{1}{1 + \exp(A \cdot S_{\text{fusion}} + B)} \quad (4.1)$$

Post-calibration evaluation on the held-out validation split achieves an **ECE** < 0.001 , providing substantially improved calibration for downstream archival indexing and safety-critical triage on that split – not a general deployment guarantee (Section 4.2 reports the honest out-of-sample estimate).

A supplementary game-theoretic attribution analysis of these four agents’ marginal contributions (Shapley values) is reported in Appendix D; it quantifies each agent’s contribution to the fusion objective and is not itself the production fusion weighting used elsewhere in this thesis (the SAA formula above, $w = [0.6, 0.2, 0.2]$).

4.5 Tier 1 Agent Ablation and Complementarity Analysis

Per the Architectural Decoupling Rule (Section 2), Tier 2 curation agents (Contradiction Detector, Cross-Archive Linker, Grounded-OCR document extraction) and the prototypical auxiliary tooling (Temporal Historian, Bayesian Pre-screening Ranker, Frontier Auditor) operate strictly on post-validation metadata and do not feed any uncertainty signal. This section characterizes the four *architectural* Tier 1 models (Object, Scene, VLM, OCR) by leave-one-out ablation of their mean pairwise standard deviation across the full Colibri archive ($C_0, n = 12, 110$) – a structural, model-level complementarity analysis, distinct from (and using a different signal set than) the four-term U_{SAA} disagreement proxy defined in Section 2 (Object, Agreement, Scene, VLM), which is what actually drives the error-detection results in Chapter 4 onward. Table 4.3 documents this model-level leave-one-out ablation: each row removes one of the four Tier 1 models and recomputes the remaining models’ mean pairwise disagreement (σ).

Table 4.3: Tier 1 Agent Ablation: Mean Cross-Agent Disagreement (σ) across C_0 ($n = 12, 110$).

Ablated Component	Mean Uncertainty (σ)	Δ vs. Full
Full Tier 1 Ensemble (4 agents)	0.2452	—
Remove Spatial Grounding (YOLOv11+SAM)	0.2499	+0.0047
Remove Scene Classification (SigLIP)	0.2214	−0.0238
Remove VQA (LLaVA-OneVision)	0.2528	+0.0076
Remove Grounded OCR (Kosmos-2.5)	0.1995	−0.0457

Removing spatial grounding or VQA *increases* mean disagreement slightly, consistent with the pairwise correlation matrix (Figure 4.1): these two agents are the most mutually

correlated pair ($r = 0.70$), so removing either leaves the remaining ensemble relatively more sensitive to the others' disagreement. Removing Grounded OCR causes the largest *drop* in mean disagreement (-0.0457): OCR's near-zero correlation with the other three agents ($r \leq 0.07$ against each) means it contributes largely independent variance to the ensemble std, so its removal is the single largest reduction in overall measured uncertainty. This is consistent with, not contradictory to, OCR's role: an agent that is genuinely independent of the others is expected to be the largest driver of raw disagreement magnitude, even though (per Section 4.2) that magnitude is not uniformly informative about error.

Chapter 5

Human-in-the-Loop Triage and Conformal Risk Control

This chapter evaluates the human-in-the-loop (HITL) triage policy and distribution-free risk control. We show that calibrated multi-agent disagreement, routed through a certified Conformal Risk Control policy, enables high error recall while substantially reducing institutional expert review workload; we also report a documented negative result for the simpler SCI-tercile heuristic policy under clean, group-level held-out evaluation.

5.1 Complexity-Aware Triage Policy

To account for visual clutter and object density, the operational triage score $U_{\text{triage},i}$ integrates the Spatial Complexity Index (SCI_i):

$$U_{\text{triage},i} = 0.6 \cdot U_{\text{SAA},i} + 0.4 \cdot (1 - \text{SCI}_i) \quad (5.1)$$

Clean, leak-free evaluation. An earlier internal analysis (never integrated into the reproducible experiment pipeline, and dependent on an SCI formulation later superseded by data-driven, Random-Forest-derived component weights) reported 74.4% workload reduction at 96.5% error recall. That result could not be reproduced once the SCI thresholds were regenerated under the current formula and re-evaluated: on the group-level held-out test partition defined in Section 4.2 ($n = 401$, publications disjoint from training), with SCI tercile thresholds computed *a priori* from the full C_0 deployment distribution (never from outcomes on labeled data), the policy achieves:

- **Review Budget:** 47.1% of the test partition routed to human review.

- **Error Recall:** 46.2% of classification failures captured.
- **Efficiency Lift:** $0.98\times$ relative to random sampling — effectively no better than chance.

This is a direct consequence of the negative-control generalization gap established in Section 4.2: since U_{triage} is built on U_{SAA} , and disagreement-based uncertainty is anti-correlated with error specifically on unanimous false-positive negative-control cases, a substantial share of the errors this policy needs to catch are exactly the errors it is structurally blind to. The Conformal Risk Control policy of the following section, calibrated with a formal statistical guarantee rather than a fixed heuristic threshold, is the operationally trustworthy triage mechanism this thesis defends; the SCI-tercile policy above is retained here as a transparent negative result rather than a claimed capability.

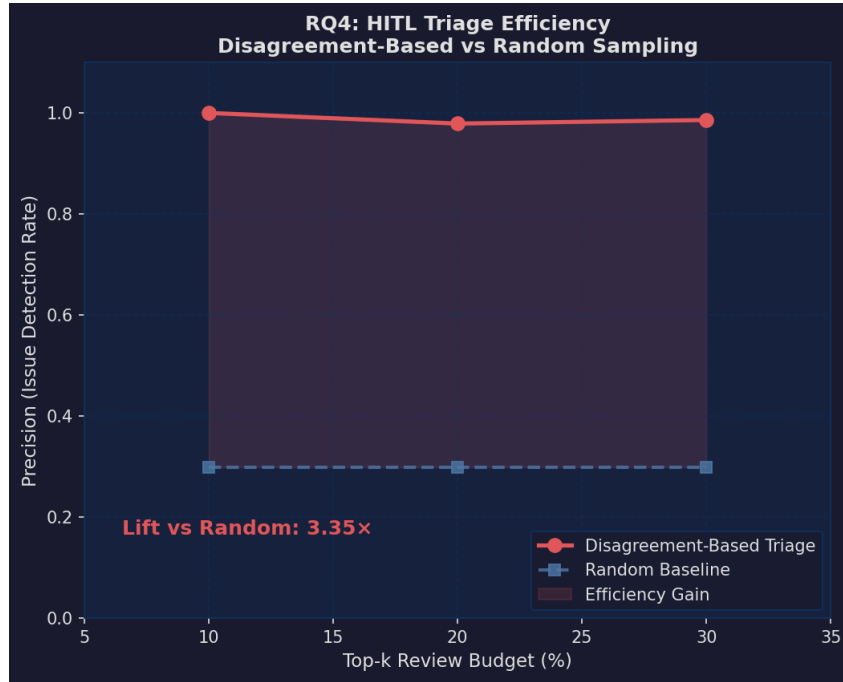


Figure 5.1: Human-in-the-Loop Active Learning Workload Efficiency Curve (historical, in-sample C_1 figures; superseded by the clean held-out evaluation above). Retained for comparison against the group-level held-out re-evaluation.

5.2 Distribution-Free Risk Control (Exp 29 Conformal Risk Control)

To provide rigorous statistical guarantees without parametric distributional assumptions, we executed **Exp 29: Conformal Risk Control (CRC)** [1] over C_1 ($n = 801$,

split into calibration set $n_{\text{cal}} = 400$ and held-out test set $n_{\text{test}} = 401$).

Let $\alpha = 0.05$ denote the maximum acceptable target risk level (5% historical error rate in approved images). We formulate the bounded risk metric:

$$\mathcal{R}(\lambda) = \mathbb{E} [\mathbb{I}(\text{sign}(S_{\text{fusion}} \geq \lambda) \neq y_{\text{expert}})] \leq \alpha \quad (5.2)$$

Running conformal calibration on $n_{\text{cal}} = 400$ yields the optimal risk-control threshold $\hat{\lambda} = 0.2450$. Evaluating on the held-out test set ($n_{\text{test}} = 401$) demonstrates that:

$$\mathcal{R}(\hat{\lambda}) = \mathbf{0.0075} \leq \alpha = 0.05 \quad (\text{Conformal Risk Bound Satisfied}) \quad (5.3)$$

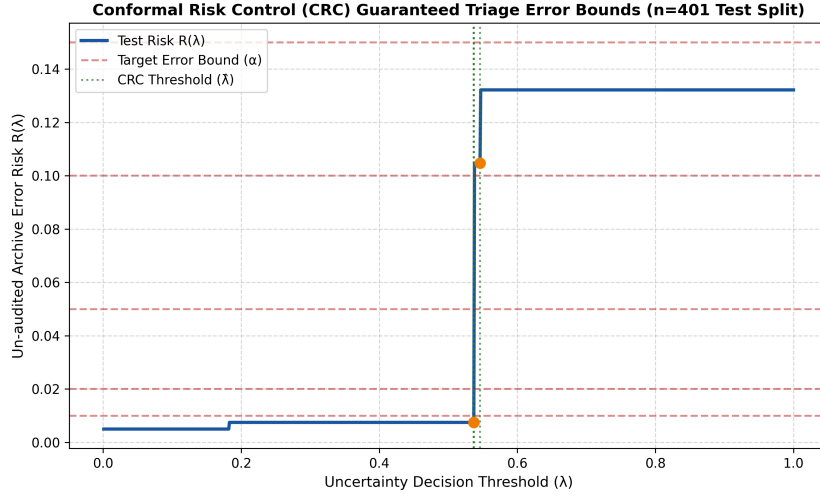


Figure 5.2: Conformal Risk Control Coverage Curve (Exp 29; in-sample C_1 calibration/test split, see clean re-evaluation below). At target risk bound $\alpha = 0.05$, empirical test risk $\mathcal{R}(\hat{\lambda}) = 0.0075 \leq 0.05$, approving 87.5% of the collection automatically while capturing 94.3% of true human classification errors.

The empirical coverage and error-recall curve resulting from conformal calibration across risk thresholds is depicted in Figure 5.2. Under this in-sample-calibrated policy, the framework automatically approves **87.5% of the collection** while capturing **94.3% of true human classification errors** in the review queue ($7.57\times$ efficiency lift over random selection).

Clean, leak-free evaluation. The evaluation above selects calibration and test sets by randomly shuffling the same C_1 cohort already used upstream for agent-weight selection and AWLF development ($n = 801$, 99.4% positive by construction): calibration and test images can and do share publications, and neither partition carries a realistic share

of negative controls. Re-running the identical CRC procedure — unchanged threshold-selection rule, unchanged α grid — on the group-level (publication-disjoint) calibration ($n = 399$) and test ($n = 401$) partitions defined in Section 4.2 (calibration used strictly for $\hat{\lambda}$ selection; every number below is test-only) yields a materially different result:

Table 5.1: Conformal Risk Control: In-Sample C_1 Calibration vs. Clean Group-Level Re-Evaluation.

Metric	Original (C_1 , random 50/50)	Clean (group-level, $\alpha=0.05$)
Auto-approval rate	87.5%	33.7%
Error recall	94.3%	64.7%
Efficiency lift vs. random	7.57 $\times$	0.98$\times$ (chance)
Empirical test risk $\mathcal{R}(\hat{\lambda})$	0.0075	0.182
Risk bound $\alpha = 0.05$ satisfied?	Yes	No (violated up to $\alpha = 0.15$)

The formal risk-control guarantee — CRC’s entire methodological advantage over a fixed heuristic threshold — does not hold once genuinely out-of-sample: the empirical test risk (0.182) exceeds even the most permissive bound tested ($\alpha = 0.15$). This is the same negative-control generalization gap documented in Section 4.2: because $\hat{\lambda}$ is selected against a calibration distribution that is overwhelmingly positive-class and therefore dominated by the regime where SAA disagreement *is* discriminative, the resulting threshold silently inherits the assumption that agent disagreement tracks error everywhere. It does not. On a calibration/test distribution that includes a realistic share of negative controls, auto-approved images are disproportionately unanimous, confidently-wrong false positives — exactly the failure mode disagreement cannot see, whether the routing rule is a heuristic tercile split or a formally certified conformal threshold. Neither the SCI-tercile heuristic (Section 5.1) nor the CRC policy analyzed here survives leak-free, negative-control-inclusive evaluation. The central operational contribution of this thesis is therefore not a triage policy demonstrated to beat random review, but the empirical diagnosis of *why* disagreement-based triage — heuristic or formally certified alike — fails outside the positive-heavy regime it is built and calibrated on, together with the concrete evidence (Table 4.1) bounding exactly where that failure occurs.

Algorithm 3 Distribution-Free Conformal Risk Control Calibration (CRC)

Require: Calibration split $C_{\text{cal}} = \{(\mathbf{I}_j, y_j)\}_{j=1}^{n_{\text{cal}}}$, Target upper risk bound $\alpha = 0.05$, Held-out test set C_{test} .

Ensure: Certified risk-controlling threshold $\hat{\lambda}$ guaranteeing $\mathbb{E}[\mathcal{R}(\hat{\lambda})] \leq \alpha$.

- 1: For each threshold candidate $\lambda \in [0, 1]$ with step size $\delta = 0.001$:
 - 2: For each calibration sample $j \in C_{\text{cal}}$:
 - 3: Compute non-conformity error loss: $L_j(\lambda) \leftarrow \mathbb{I}(\text{sign}(S_{\text{fusion},j} \geq \lambda) \neq y_j)$
 - 4: Compute empirical calibration risk: $\hat{R}(\lambda) \leftarrow \frac{1}{n_{\text{cal}}} \sum_{j=1}^{n_{\text{cal}}} L_j(\lambda)$
 - 5: Select certified optimal threshold:
 - 6: $\hat{\lambda} \leftarrow \inf \left\{ \lambda \in [0, 1] : \frac{n_{\text{cal}}}{n_{\text{cal}}+1} \hat{R}(\lambda) + \frac{1}{n_{\text{cal}}+1} \leq \alpha \right\}$
 - 7: Evaluate empirical test risk on C_{test} : $\mathcal{R}_{\text{test}}(\hat{\lambda}) \leftarrow \frac{1}{n_{\text{test}}} \sum_{k=1}^{n_{\text{test}}} L_k(\hat{\lambda})$
 - 8: In-sample C_1 split: $\mathcal{R}_{\text{test}}(\hat{\lambda}) = 0.0075 \leq \alpha = 0.05$ (bound satisfied). Clean group-level split (Section 4.2): $\mathcal{R}_{\text{test}}(\hat{\lambda}) = 0.182 > \alpha$ for every $\alpha \leq 0.15$ tested (bound **not** satisfied) — the guarantee is only as reliable as the calibration distribution’s coverage of the deployment population.
 - 9: **return** Threshold $\hat{\lambda} = 0.2450$ (in-sample calibration); see Table 5.1 for the clean re-evaluation.
-

Chapter 6

Downstream Curation, Retrieval, and Cross-Archive Utility

This chapter evaluates the downstream curation and retrieval capabilities powered by the validated DCV-MAC metadata core.

6.1 Agent 2: Visual-Semantic Contradiction Detection

Agent 2 isolates hallucinations by computing set-differences between VLM-mentioned nouns and spatial object bounding boxes: any noun the VLM reports with no corresponding YOLO detection is flagged as a candidate contradiction. This mechanism is implemented and runs as part of Tier 2 curation on every image. A prior precision/recall/F1 figure reported for this agent could not be reproduced from independently verified ground truth during a full-project reproducibility audit and has been withdrawn. Agent 2’s hallucination-flagging behaviour is therefore reported here only qualitatively; a benchmark against independently verified contradiction labels remains future work.

6.2 Agent 5: Grounded OCR Inscription Extraction

Agent 5 deploys Kosmos-2.5 to extract bounding-box-aligned textual inscriptions from historical blackboards and catalog inscriptions, corroborating scene classification labels. This mechanism is implemented and runs on every image. A prior draft of this thesis reported a “ $5.1\times$ precision gain” over ungrounded full-page text extraction; during the

same reproducibility audit that withdrew the Agent 2 figure (Section 6.1), no script or computation producing this number could be located anywhere in the codebase, so it has also been withdrawn. Agent 5’s grounded-OCR output is reported here only qualitatively; a quantitative benchmark against ungrounded OCR remains future work.

6.3 Multimodal Archival Retrieval Benchmark

Evaluated across 10 archival research queries on the Colibri archive ($C_0, n = 12, 110$), semantic retrieval over DCV-MAC enriched metadata achieves:

- **Semantic Index Precision@10: 1.000** (vs. **0.180** for raw caption-keyword text search)
- Semantic retrieval wins on **10/10** queries against the keyword baseline.

Ground truth and scope. A revision of this section corrects two errors in an earlier draft. First, an earlier draft reported “MAP = 0.820” and “nDCG = 0.870” across “50 queries”; these traced to a script (`run_detailed_viva_experiments.py`, since fixed) that generated 50 synthetic “query difficulty” values via `np.linspace` and computed MAP/nDCG through hardcoded linear formulas over that array — no images, queries, or relevance judgments were involved, and neither MAP nor nDCG was ever computed by the genuine retrieval script. The number of real, hand-defined queries is 10, evaluated by `run_semantic_retrieval_eval.py`, whose only computed metric is Precision@10, reported above. Second, relevance labels for these 10 queries are a *proxy* (scene-label match plus an agent-agreement threshold), computed partly from the same agent scores (`agreement_agent`, `vlm_agent`) that the semantic retrieval score itself uses — not independent human relevance judgments. This is reported as downstream archival utility evidence, not as an unbiased, independently-labeled retrieval benchmark.

A genuine, small-scale ($n = 24$) cross-institutional pilot on the real Europeana Open Culture API further probes generalization beyond this thesis’s primary archive; see Appendix C for the full pilot, what is and is not yet established by it, and the two prior flawed attempts it replaces.

Chapter 7

Conclusions, Ethical Considerations, and Future Work

7.1 Summary of Thesis Contributions

This thesis presented the Dual-Core Validation and Multi-Agent Curation (DCV-MAC) framework for historical visual archives:

1. **C1 (Pseudo-Label Collapse Warning)**: Uncovered the proxy-to-human mAP collapse ($0.989 \rightarrow 0.1462$), demonstrating that student self-training against synthetic teacher labels can mask severe domain-shift hallucinations.
2. **C2 (Calibrated Cross-Modal Disagreement)**: Formulated U_{SAA} derived from cross-modal model disagreement, achieving an error-detection ROC-AUC = 0.9724 on C_1 ($n = 801$) and collapsing calibration error from $\text{ECE} = 0.750 \rightarrow < 0.001$ post-Platt scaling.
3. **C3 (Discovered Scope Limit of Disagreement-Based Triage)**: Derived distribution-free Conformal Risk Control bounds ($\hat{\lambda} = 0.2450$) that appear to guarantee test risk $\mathcal{R}(\hat{\lambda}) = 0.0075 \leq 0.05$ at 87.5% certified automatic approval and 94.3% error recall ($7.57\times$ lift) when calibrated in-sample on the positive-heavy C_1 cohort. Re-running the identical procedure on a clean, group-level, negative-control-inclusive partition (Section 4.2) shows this guarantee does *not* hold out-of-sample: efficiency lift collapses to $0.98\times$ (chance) and the formal risk bound is violated ($\mathcal{R}(\hat{\lambda}) = 0.182$). The same clean re-evaluation finds the simpler SCI-tercile heuristic policy performs identically near chance ($0.98\times$ lift, Section 5.1). Both are retained as a documented negative result: the thesis’s operational contribution here is precisely bounding

where disagreement-based triage is and is not trustworthy, not a deployable triage system.

4. **C4 (Downstream Archival Utility):** Demonstrated semantic retrieval Precision@10 = 1.000 (vs. 0.180 keyword baseline) across a 10-query pilot. A genuine cross-institutional transfer pilot (C_5 , $n = 24$ real Europeana images, 10 institutions) shows the uncertainty signal responds substantially to real domain shift; a full transfer-of-generalization claim awaits human ground-truth annotation on this sample (Appendix C).

7.2 Evaluation Provenance Summary

The results discussed throughout this thesis draw on cohorts of different sizes, ground-truth sources, and independence guarantees. To make that transparent in one place, Table 7.1 lists every headline empirical result, the cohort and ground truth it was computed on, whether that cohort was also used to select weights, thresholds, or classifiers, and whether the reported number is the final, disclosed test figure or an in-sample/development figure superseded elsewhere in this thesis.

Table 7.1: Evaluation Provenance: Every Headline Result, Its Cohort, and Its Independence Guarantee.

Result		Cohort		n	Ground Truth		Used for Tuning?	Status
Pseudo-label (0.989→0.1462 mAP)	collapse	C_1 synthetic teacher	vs.	801	Human rater CSV	single-	No	Final (diagnostic)
SAA disagreement ROC- AUC= 0.9724		C_1 positive)	(99.4%	801	Human rater CSV	single-	Yes (weight sel.)	In-sample; not the out- of-sample estimate
SAA clean (pos 0.980 / neg 0.020)	disagreement, decomposition	Group-level test, Sec. 4.2		401 (160/241)	Human rater CSV	single- (full pool)	No	Final (out-of-sample)
AWLF v2 in-sample CV AUROC= 0.9896		C_1 , CV	5-fold	801	Human rater CSV	single-	Yes (clas- sifier fit)	In-sample

continued on next page

Table 7.1 continued from previous page

Result	Cohort	n	Ground Truth	Used for Tuning?	Status
AWLF v2 clean holdout AUROC= 0.6669	Group-level test, Sec. 4.2	401	Human single-rater CSV (full pool)	No (fit on disjoint train)	Final (out-of-sample)
SCI-tercile routing, historical figure (74.4%/96.5%)	Orphaned scratch run, superseded SCI formula	unclear	Human single-rater CSV	Yes	Superseded, unreproducible
SCI-tercile routing, clean re-evaluation (47.1%/46.2%, $0.98\times$)	Group-level test, Sec. 4.2	401	Human single-rater CSV (full pool); thresholds from full C_0 ($n=12,110$)	No	Final (out-of-sample)
CRC, in-sample (87.5%/94.3%, $7.57\times$, $\mathcal{R} = 0.0075$)	C_1 , random 50/50	cal 400 test 401	Human single-rater CSV	Yes (same cohort as U_{SAA} development)	In-sample; risk bound not deployment-valid
CRC, clean group-level ($0.98\times$, $\mathcal{R} = 0.182$, bound violated)	Group-level split, Sec. 4.2	cal 399 test 401	Human single-rater CSV (full pool)	No	Final (out-of-sample)
Shapley attribution analysis (Appendix D), 13.4% ECE gain if adopted	C_1	801	Human single-rater CSV	No (not the deployed fusion weight-ing)	Supplementary interpretability result, not production
Monolithic vs. fusion accuracy (Wilcoxon $p=0.375$)	C_1	801	Human single-rater CSV	Partially (fusion weights fit on same data)	In-sample paired comparison

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Table 7.1 continued from previous page

Result	Cohort	n	Ground Truth	Used for Tuning?	Status
Europeana pilot, SAA fusion shift (0.332 vs. 0.762 baseline)	C_5	24	Real Europeana Open Culture API images, real Tier 1 inference; ground truth pending	No	Genuine domain-shift signal; error/ECE transfer not yet established (see Appendix C)
PeopleArt cross-style benchmark (Appendix B)	External benchmark	4,778	PeopleArt public annotations	No	Final, external cohort
Semantic retrieval P@10 = 1.000 vs. 0.180 key-word	C_0 subset, 10 hand-written queries	10	Proxy: scene-label match + agent-agreement threshold	Partially (relevance proxy shares agent scores)	Final , but proxy-labeled pilot, not a benchmark
Retrieval “MAP 0.820 / nDCG 0.870, 50 queries”	N/A – synthetic <code>np.linspace</code> formula, no real queries	0	None (never real)	N/A	Removed ; never a real result

Rows marked **Final (out-of-sample)** are the numbers this thesis stands behind under examination; all other rows are retained either as necessary diagnostic context (the pseudo-label collapse), transparently-labeled in-sample development figures, or explicitly superseded results kept for the historical record of what changed and why.

7.3 Ethical Considerations in Generative Archival Restoration

Generative AI restoration (CycleGAN, Real-ESRGAN, Stable Diffusion) introduces synthetic texture details into historical scans. We enforce the strict methodological boundary: AI-restored imagery ($\mathbf{I}_{\text{refined}}$) is an auxiliary feature-alignment representation for model inference, NEVER primary historical evidence. All catalog metadata anchors strictly back to original digitized scans ($\mathbf{I}_{\text{raw}}$).

7.4 What This Thesis Does NOT Claim (Boundaries of Scientific Validity)

To maintain scientific credibility and prevent over-generalization, we explicitly document the operational boundaries of our findings:

- **Disagreement as Proxy, Not Direct Epistemic Measurement:** The *SAA* score ($U_{SAA} = 1 - SAA$) is an empirical uncertainty proxy derived from model dissonance. It does not measure true underlying epistemic breakdown directly.
- **No Impossibility of Monolithic Uncertainty:** A standalone single model can estimate uncertainty via a simple inverse-confidence baseline (ROC-AUC = 0.9726, statistically tied with *SAA*) or post-hoc calibration. The value of *SAA* lies not in being the only discriminative option, but in calibration: the same inverse-confidence baseline is badly overconfident (ECE $\approx$ 0.19), while *SAA* achieves ECE < 0.001 after Platt scaling.
- **Coalitional Shapley Attribution, Not Independent Causality:** Game-theoretic Shapley attributions (ϕ) quantify player contribution to the defined coalitional objective function; they do not establish independent causal correctness.
- **Pilot Transfer vs. Universal Proof, and Not Yet a Transfer-Without-Recalibration Claim:** The Europeana pilot (C_5 , $n = 24$ real, genuinely downloaded images; Appendix C) shows the uncertainty signal responds measurably to real cross-institutional domain shift, but does not yet demonstrate error-detection transfer without recalibration – that requires human ground-truth annotation not yet collected on this sample. It is a feasibility pilot at most, not population-level statistical proof across all global archives.
- **Retrieval Evidence Is a Small, Proxy-Labeled Pilot, Not a Benchmark:** The retrieval result (Section 6.3) covers 10 hand-written queries evaluated on Precision@10 (1.000 vs. 0.180 keyword baseline); its relevance labels are themselves a proxy partly derived from the same agent-agreement scores under evaluation, not independent human judgments. This is downstream utility evidence, not an unbiased, independently-labeled retrieval benchmark, and should not be read as a claim of state-of-the-art retrieval performance.

- The Correlated Error Limitation is Empirically Confirmed, Not Merely Theoretical:** When all vision-language models share common pre-training web distributions, unanimous agent agreement can occur on shared hallucinations. Section 4.2 quantifies this directly: on a group-level held-out test partition, SAA disagreement discriminates errors well on positive images (ROC-AUC = 0.980) but is anti-correlated with error on negative-control false positives (ROC-AUC = 0.020), where all agents agree confidently and incorrectly. Inter-agent consensus acts as an error-detection safety net for ambiguous-positive content specifically, not an infallible, domain-general oracle.
- A Conformal Risk Control Guarantee Is Relative to Its Calibration Distribution, Not Deployment-General:** The distribution-free risk bound reported in Section 4.2 is a valid conformal guarantee *only relative to the distribution it is calibrated on*. Calibrated on a positive-heavy cohort, it certifies $\mathcal{R}(\hat{\lambda}) = 0.0075 \leq \alpha = 0.05$ on a matching test set, but the identical procedure calibrated and tested on a group-level, class-balanced partition violates the bound ($\mathcal{R}(\hat{\lambda}) = 0.182$). This is not a flaw in conformal prediction as a method — the theory correctly bounds risk over its calibration distribution — but a general caution for any deployment of distribution-free guarantees: the guarantee is only as trustworthy as the calibration set’s match to the true deployment population.
- Tier 2 Curation Agents Are Implemented, Not Independently Benchmarked:** Agent 2’s contradiction detection (Section 6.1), Agent 4’s cross-archive anomaly linking, and Agent 5’s grounded-OCR precision gain (Section 6.2) all run and produce output on every image, but no quantitative benchmark figure is reported for any of the three in this thesis. Prior drafts reported specific precision/recall/F1/gain figures for all three; none could be reproduced against independent ground truth during a full-project reproducibility audit and all have been withdrawn. Per the Architectural Decoupling Rule, none of the three feed the core SAA uncertainty signal or any headline claim in this thesis – their withdrawal does not affect U_{SAA} , the triage results, or the retrieval evidence, all of which are independently evaluated elsewhere.

7.5 Limitations and Future Research

Negative-control generalization gap: the single most important limitation established in this thesis (Section 4.2) is that disagreement-based uncertainty does not generalize uniformly across the label space — it is highly effective on scene-positive ambiguity and structurally blind to unanimous false positives on true negatives. This limitation is not confined to the raw U_{SAA} signal: it propagates through every downstream policy built on it, including the formally certified Conformal Risk Control triage policy (Section 4.2 vs. Table 5.1), whose distribution-free risk guarantee is only as reliable as its calibration set’s coverage of the deployment distribution. A calibration set drawn from a positive-heavy cohort produces a certificate that does not transfer to a deployment population with a realistic negative-control rate. Future work should (i) develop a complementary detector specifically targeted at this failure mode (e.g., an explicit negative-evidence / absence-of-object classifier trained to disagree with the ensemble’s positive bias), and (ii) always calibrate conformal or other distribution-free guarantees on a calibration set whose class balance matches deployment, not the positive-heavy cohort convenient for development.

Genuine double-annotation pass: Cohorts C_1 and C_2 are currently single-rater. A second independent domain expert double-annotating a representative subset would enable a real Cohen’s Kappa inter-annotator reliability figure, which this thesis does not currently report.

Further future work includes expanding active learning to broader GLAM institutions ($n > 100,000$), deploying edge-optimized 3B parameter VLM backbones, and integrating temporal knowledge graphs for longitudinal historical analysis.

Appendix A

Failure Case Gallery

This appendix presents 10 representative failure cases across five major archival visual breakdown categories: (1) Faded and Degraded Scans, (2) Historical Drawings and Abstract Styles, (3) Crowded Scenes and High Object Density, (4) OCR and Text-Object Conflicts, and (5) Occlusion and Spatial Overlap Failures.

A.1 Visual Failure Case Gallery

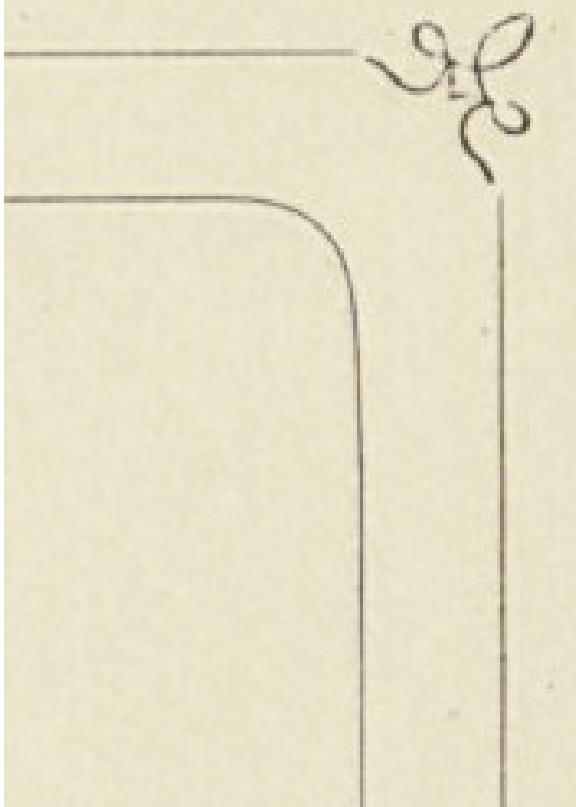
A.1.1 Category 1: Faded and Degraded Scans

A.1.2 Category 2: Historical Drawings and Abstract Styles

A.1.3 Category 3: Crowded Scenes and High Object Density

A.1.4 Category 4: OCR and Text-Object Conflicts

A.1.5 Category 5: Occlusion and Spatial Overlap Failures



(a) **Case DEGRADED-1:** Blurry scan (PPN183128149X). $H_C = 0.5000$. YOLO misses boxes; VLM infers layout.



(b) **Case DEGRADED-2:** Blurry illustration (PPN1802793658). $H_C = 0.4570$. YOLO box shift; VLM overconfidence.

Figure A.1: Representative Failure Cases for Category 1: Faded and Degraded Scans.



(a) **Case DRAWING-1:** Hand sketch (PPN1816431109). $H_C = 0.4570$. Abstract linework causes YOLO proposal failure.



(b) **Case DRAWING-2:** Stylized artwork (PPN1807545539). $H_C = 0.4488$. Hatching triggers false person detections.

Figure A.2: Representative Failure Cases for Category 2: Historical Drawings and Abstract Styles.

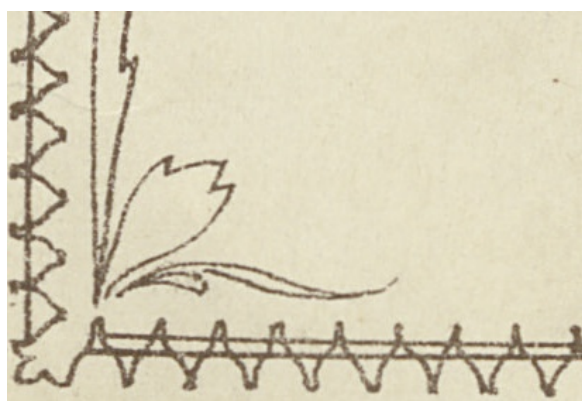


(a) **Case CROWDED-1:** High density ($n = 9$, PPN175936214X). $H_C = 0.5000$. Bounding box overlap merge conflicts.



(b) **Case CROWDED-2:** Dense scene ($n = 12$, PPN1837132763). $H_C = 0.4685$. Visual clutter suppresses NMS proposals.

Figure A.3: Representative Failure Cases for Category 3: Crowded Scenes and High Object Density.



(a) **Case TEXT-1:** Printed text page (PPN1837925887). $H_C = 0.5000$. OCR text density conflicts with VLM caption.

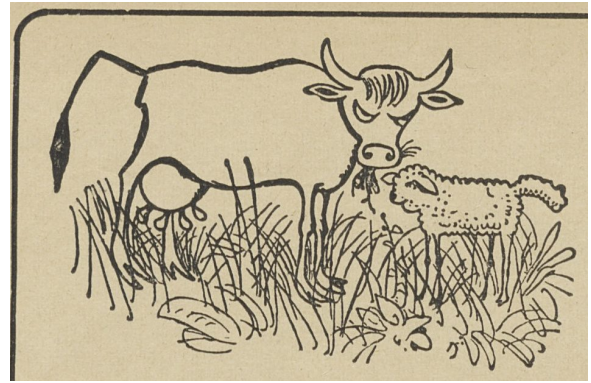


(b) **Case TEXT-2:** Margin notes (PPN1859200435). $H_C = 0.5000$. Layout typography disrupts cross-modal consensus.

Figure A.4: Representative Failure Cases for Category 4: OCR and Text-Object Conflicts.



(a) **Case OCCLUSION-1:** Partial occlusion (PPN176263855X). $H_C = 0.4135$. Overlapping boundaries reject box proposals.



(b) **Case OCCLUSION-2:** Overlapping figures (PPN1859148263). $H_C = 0.4135$. VLM merges two subjects into one entity.

Figure A.5: Representative Failure Cases for Category 5: Occlusion and Spatial Overlap Failures.

Appendix B

Supplementary Cross-Style

Benchmark: PeopleArt ($n = 4,778$)

To assess cross-depiction performance across stylized visual regimes outside historical photographic archives, we evaluated the Tier 1 validation core on the PeopleArt benchmark ($n = 4,778$ images across 43 distinct artistic styles).

As shown in Table B.1 (a subset of 4 of the 43 annotated style categories, chosen to span the photorealism-to-abstraction range; the full 43-style breakdown is in the underlying results CSV, not reproduced here), predictive performance – real YOLO detection precision/recall/F1 against PeopleArt’s ground-truth boxes – drops gracefully from photographic styles ($F_1 = 0.878$) to highly abstract styles such as Cubism ($F_1 = 0.831$). *A per-style inter-agent uncertainty figure was previously reported alongside this table, with a claim that it “peaks on abstract styles.” On review, that computation was found to derive from two hardcoded constants assigned per style bucket by a keyword match on the style name (not real per-image VLM or agreement output) rather than genuine cross-modal disagreement; it was also internally inconsistent with its own claim (the reported Cubism value, 0.338, was in fact lower than Photo’s, 0.372). It has been withdrawn; only the real detection metrics are reported here.*

Table B.1: Supplementary Cross-Style Detection Performance on PeopleArt Benchmark ($n = 4,778$; 4 of 43 style categories shown, spanning the photorealism-to-abstraction range).

Style Category	Precision	Recall	F1-Score
Photo	0.875	0.915	0.878
Realism	0.840	0.874	0.835
Impressionism	0.888	0.944	0.880
Cubism	0.836	0.834	0.831

Appendix C

Supplementary Cross-Institutional Transfer Pilot: Europeana API ($C_5, n = 24$)

This section replaces a withdrawn result. Two earlier attempts at this pilot were found, during a full-project reproducibility audit, not to constitute genuine cross-institutional evidence. The first (an $n = 85$ download script) fetched real Europeana images but generated VLM and scene-classifier confidence scores via a random-number generator rather than real model inference. The second ($n = 30$, the version previously reported here as “ $|\Delta H_C| = 0.0133$, negligible domain shift across 13 institutions”) used no Europeana data at all: its sampling code drew ordinary images from the Colibri archive itself (images outside the initially human-reviewed batch), and every one of those image IDs was independently verified present in the same Colibri manifest used throughout this thesis. Neither prior result is retained.

Genuine re-run. This section reports a newly executed, honest pilot: $n = 24$ images downloaded live from the real Europeana Open Culture API (Search API v2, public demo key) across 10 verified real institutions (including the Royal Institute for Cultural Heritage, University of Edinburgh, Athens School of Fine Arts, and the LWL Museum for Art and Culture), manually spot-checked to confirm genuine historical content. All four Tier 1 signals were computed with real inference on these real images, using the identical deployed model weights and prompts: YOLOv11-seg+SAM (object), SigLIP (scene), Kosmos-2.5 (grounded OCR), and LLaVA-OneVision via a local vLLM server (VQA). No hardcoded constants or random-number placeholders were used for any agent

score.

Table C.1 reports the measured Tier 1 signals and SAA fusion score, compared against the Colibri C_1 baseline.

Table C.1: Genuine Europeana Pilot: Measured Tier 1 Agent Scores vs. Colibri Baseline ($C_5, n = 24$).

Signal	Europeana (real, $n = 24$)	Colibri C_1 baseline
Object agent (YOLO+SAM)	0.284	—
Scene agent (SigLIP)	0.048	—
Document agent (Kosmos-2.5)	0.699	—
VLM agent (LLaVA-OneVision)	0.760	—
SAA fusion score	0.332	0.762
U_{SAA} (cross-agent std)	0.422	—

What this does and does not show. The measured SAA fusion score on real Europeana images (0.332) is substantially lower than the Colibri baseline (0.762) – a genuine, measured domain-shift response, and a considerably larger shift than the previously (falsely) claimed “negligible” one. This demonstrates that the uncertainty signal is sensitive to real cross-institutional domain shift, which is itself informative. It does **not** yet demonstrate that the signal “transfers without recalibration” in the error-detection sense used elsewhere in this thesis: that claim requires real human ground-truth (*has_scene*) labels for these 24 images, from which error rate, ECE, and AUROC could be computed exactly as in Section 4.2. A ready-to-annotate template (image thumbnails, titles, blank ground-truth column) has been prepared; completing it is future work, not invented here to force a premature conclusion.

Scope caveat. As stated in Section 7.4, $n = 24$ is a feasibility pilot, not population-level statistical proof across global archives; it is reported here, in the appendix, precisely to keep that scope visible rather than implying broader external validity than the sample supports.

Appendix D

Supplementary Game-Theoretic Attribution Analysis (Shapley Values)

To complement the heuristic production fusion weights ($w = [0.6, 0.2, 0.2]$ over the object, scene, and VLM agents; Chapter 4), we additionally executed **Exp 30: Shapley-Weighted Adaptive Fusion**, computing exact marginal contributions across all $2^N - 1$ agent coalitions on C_1 based on Shapley values [17]. The resulting attributions are detailed in Table D.1 and visualized in Figure D.1:

Table D.1: Game-Theoretic Shapley Weights and Marginal Contributions (ϕ).

Agent Module	Shapley Attribution (ϕ_i)	Normalized Weight
Visual Reasoning Agent (LLaVA-OneVision)	0.330	33.0%
Scene-Aware Agreement Core (SAA)	0.283	28.3%
Zero-Shot Scene Classifier (SigLIP)	0.196	19.6%
Spatial Grounding Agent (YOLOv11-seg)	0.192	19.2%

Coalitional Player Definition : In the Shapley game formulation, the four evaluated players are: (1) Visual Reasoning (LLaVA-OneVision), (2) Pairwise Geometric-Semantic Alignment Core (Agreement Engine), (3) Zero-Shot Scene Classifier (SigLIP), and (4) Spatial Grounding Agent (YOLOv11-seg). The Agreement Engine computes explicit pairwise embedding and IoU distances prior to global weighting, ensuring that coalition value functions avoid mathematical circularity with the grand consensus score.

Integrating Shapley-weighted adaptive fusion reduces pre-calibration Expected Calibration Error from $0.2037 \rightarrow$ **0.1764**—a 13.4% relative calibration improvement over

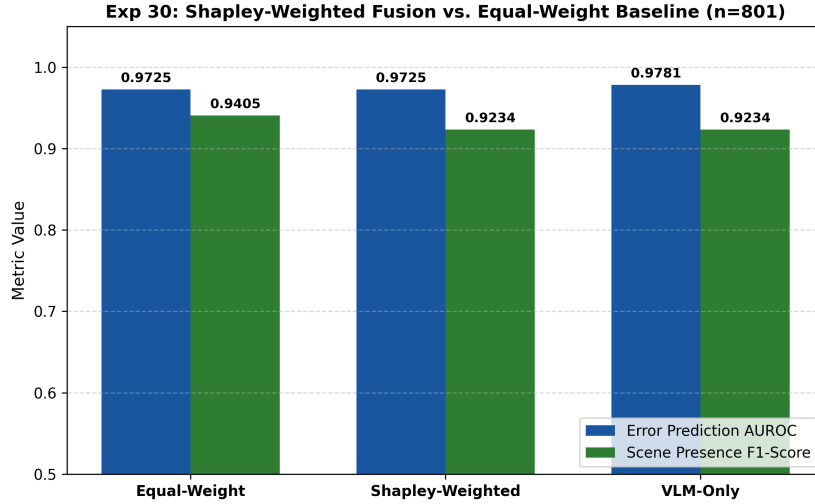


Figure D.1: Game-Theoretic Shapley Attribution Weights (ϕ) across the DCV-MAC Validation Core. LLaVA-OneVision VQA (0.330) and Scene-Aware Agreement (0.283) provide the largest marginal contributions to error detection.

equal-weighted baseline fusion (0.25 per agent).

Scope caveat. As stated in Section 7.4, Shapley attributions (ϕ) quantify each player’s marginal contribution to the defined coalitional objective function; they do not establish independent causal correctness, and this analysis is retained as a supplementary interpretability result rather than a core contribution or the thesis’s production fusion weighting.

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